GENERAL SCIENCE GRADE 10: CHEMICAL FAMILIES

CBE Phenomenon-Driven Lesson Sequence — Sub-Strand 2.2 (8 Lessons)

SUB-STRAND OVERVIEW	
Grade Level	10
Subject	General Science
Strand	Strand 2.0: Matter and Chemical Reactions
Sub-Strand	Sub-Strand 2.2: Chemical Families
Total Duration	8 lessons × 40 minutes = 320 minutes total
Sub-Strand Content	– Alkali metals — properties, reactions with air, water and dilute acids; examples: sodium (Na), potassium (K), lithium (Li) – Alkaline earth metals — properties, reactions with air, water and dilute acids; examples: magnesium (Mg), calcium (Ca) – Halogens — physical and chemical properties; examples: chlorine (Cl), bromine (Br), iodine (I), fluorine (F) – Noble gases — unreactive nature and day-to-day applications; examples: helium (He), neon (Ne), argon (Ar) – Transition metals — selected properties and uses; examples: copper (Cu), iron (Fe), zinc (Zn), lead (Pb) – Physical properties of selected elements from each chemical family (state, colour, lustre, density, melting/boiling point, conductivity) – Chemical properties of selected elements — reactions of sodium, magnesium and chlorine with air (oxygen), water and dilute acids – Balanced chemical equations for reactions of elements in each family – Uses of elements and their compounds in day-to-day life (road illumination, water treatment, food preservation, construction, medicine) – Safety precautions and first-aid procedures when handling reactive elements in the laboratory
Learning Outcomes	By the end of the sub-strand, the learner should be able to: a) identify chemical families in the periodic table, b) classify elements of the periodic table into their chemical families, c) investigate the physical and chemical properties of selected elements of the periodic table, d) outline the uses of elements of the periodic table, e) appreciate the uses of the elements of the periodic table in day-to-day life.
Core Competencies	– Self-efficacy: as the learner carries out activities to investigate the reaction of sodium, magnesium and chlorine with air and dilute acids, building confidence in experimental science. – Digital literacy: as the learner uses available digital and print media to research on transition elements and their uses, and shares findings using digital presentation modes. – Communication and collaboration: as the learner sorts, groups and discusses properties of chemical families with peers, and presents conclusions in plenary. – Critical thinking and problem solving: as the learner analyses patterns in physical and chemical properties across families to explain why certain elements are suited to specific uses.

Core Values	– Love: as the learner takes care of self and others by observing safety when carrying out activities to investigate properties of elements of the periodic table and appropriately using first aid kits in case of injuries. – Responsibility: as the learner handles reactive elements and digital devices carefully, following laboratory protocols and ethical research practices. – Respect: as the learner recognises and appreciates the contributions of every group member during discussions and practical investigations. – Patriotism: as the learner connects the properties and uses of elements to Kenya's industrial, agricultural and infrastructural development.
Science & Engineering Practices	– Planning and Carrying Out Investigations: as the learner designs and conducts experiments on the physical and chemical properties of alkali metals, alkaline earth metals and halogens, controlling variables and recording systematic observations. – Analysing and Interpreting Data: as the learner compares experimental results across chemical families to identify trends in reactivity and physical properties. – Constructing Explanations: as the learner uses evidence from investigations and research to explain why elements within the same family share similar properties and have related uses. – Obtaining, Evaluating and Communicating Information: as the learner researches transition elements using digital and print media and communicates findings to peers through presentations and discussions. – Engaging in Argument from Evidence: as the learner defends classifications of elements into families using experimental evidence and data from the periodic table.
Pertinent & Contemporary Issues (PCIs)	– Road safety: as the learner discusses with peers the properties of elements that make them suitable for use in illuminating roads (e.g., neon and sodium vapour lamps on Nairobi's Thika Superhighway and Mombasa Road). – Safety and security: as the learner observes safety precautions and appropriately uses first aid kits when carrying out activities to investigate properties of reactive elements of the periodic table. – Environmental issues: as the learner examines the environmental impact of reactive elements and their compounds (e.g., chlorine in water treatment at Nairobi Water's Ruiru plant; sodium compounds in soil salinity along the Rift Valley). – Socio-economic issues: as the learner relates the properties of transition metals (iron, copper, zinc) to Kenya's construction, manufacturing and agriculture sectors.
Career Connections	– Industrial chemist: designing and optimising chemical processes using reactive alkali metals, halogens and transition metals in Kenyan manufacturing industries. – Materials engineer / metallurgist: selecting appropriate metals (iron, copper, zinc) for construction, electrical wiring and infrastructure projects such as the SGR and Nairobi Expressway. – Water treatment technologist: applying knowledge of chlorine (halogen) chemistry to purify drinking water for cities like Nairobi, Mombasa and Kisumu. – Pharmacist / medical technologist: using knowledge of chemical families to understand drug compounds, antiseptics and mineral supplements relevant to Kenyan healthcare. – Agricultural extension officer: advising Kenyan farmers in the Rift Valley and Central Kenya on soil mineral content, fertiliser chemistry and the role of trace transition metals in crop nutrition. – Electrical technician / electronics engineer: applying properties of noble gases (argon in welding, neon in signage) and conductive metals in practical Kenyan contexts.
Focus for Lessons	This unit investigates how elements of the periodic table are organised into chemical families — alkali metals, alkaline earth metals, halogens, noble gases and transition metals — and how shared electron arrangements give each family its characteristic physical and chemical properties. Learners engage in hands-on investigations, data analysis and peer discussions anchored to a puzzling Kenyan phenomenon to build an evidence-based understanding of why elements within the same family behave similarly

	and why their properties make them uniquely suited to specific everyday uses. Throughout the unit, Kenyan industrial, agricultural and infrastructural contexts are used to make abstract periodic trends tangible and personally relevant.
Driving Question / Key Inquiry Questions	DRIVING QUESTION: Why do some elements explode in water, some glow silently in glass tubes, and others rust on a Kenyan farm — and how do their "family" properties explain the materials that build, light up, and sustain our daily lives? KICD KEY INQUIRY QUESTION: 1. How are elements suited to their use in day-to-day life?
Anchoring Phenomenon	ANCHORING PHENOMENON — "The Street Lights of Thika Road Glow Orange, But the Welding Torch Burns Blue-White and Leaves No Ash" Students living near or travelling along Nairobi's Thika Superhighway notice that the tall street lamps cast a distinctive orange-yellow glow every night, yet a jua kali (informal sector) welder in Kamukunji uses a gas that produces an intensely bright, almost colourless flame that neither burns nor reacts with the metal being welded. Meanwhile, the white powder a student's parent sprinkles into the household water storage tank (calcium hypochlorite, commonly called "pool chlorine") fizzes and releases a sharp, choking smell when it contacts water — yet the noble gas inside a party balloon (helium) is completely inert and harmless. All four substances — sodium (in the street lamp), argon (in the welding shield), chlorine compounds (in the water treatment tablet), and helium (in the balloon) — are elements, yet they behave in strikingly different ways. Students are puzzled: if all these are elements made of atoms, why do some explode on contact with water, some glow in sealed tubes, some bleach and disinfect, and some do absolutely nothing? The puzzle deepens when students realise that iron (in a jembe/hoe), copper (in electrical wiring), and zinc (on a mabati/corrugated iron roof) are also elements, yet they are solid, shiny and conduct electricity — properties utterly unlike sodium or chlorine. This phenomenon anchors the entire unit: students must explain the patterns in element behaviour by uncovering the concept of chemical families.
Storyline Thread	Lesson 1 — PREDICT: "Family Resemblances" — Anchoring Phenomenon Launch Students are shown the four anchoring images/objects: a Thika Road sodium lamp, a jua kali argon-welding setup, a calcium hypochlorite water tablet fizzing in a cup, and a helium party balloon. Working in groups, students record predictions: What do these substances have in common? Why do they behave so differently? Students construct an initial class Driving Question Board (DQB) listing their wonderings. Teacher introduces the term "chemical family" and students sort the first 20 elements (cards provided) by guessing which belong together — without being told the rules — activating prior knowledge from Sub-Strand 2.1 (Periodic Table) on electron arrangements. Lesson 2 — OBSERVE & EXPLAIN: Alkali Metals — Dramatic Reactivity Students watch a teacher demonstration (or video clip) of sodium reacting with water (fizzing, movement, possible ignition) and compare it to lithium and potassium. Groups record observations and write balanced equations for Na + H_2O and K + H_2O. Students discuss: why does reactivity increase down Group 1? Teacher connects observations to electron arrangement (one outer electron, easily lost). Students update DQB: "We now know alkali metals lose one electron easily — but why do alkaline earth metals behave differently?" Lesson 3 — OBSERVE & EXPLAIN: Alkaline Earth Metals — Moderate Reactivity Students carry out a supervised experiment: magnesium ribbon reacts with dilute hydrochloric acid (HCl) and with water (cold vs. hot). Observations are recorded and balanced equations written (Mg + HCl, Mg + H_2O steam). Students compare reactivity of Group 1 vs. Group 2, linking two outer electrons to lower reactivity. Teacher connects to real Kenyan context: magnesium and calcium in fertilisers (CAN — Calcium Ammonium Nitrate) used by Rift Valley maize farmers. Students update DQB and their initial

	element-sorting cards. Lesson 4 — OBSERVE & EXPLAIN: Halogens — Reactive Non-Metals and the Bleaching Mystery Students perform the colour-changing flower experiment (supporting phenomenon 2) using chlorine water and a hibiscus flower. Groups observe bleaching action and connect to chlorine's use in Nairobi Water's purification process. Teacher demonstrates iodine's colour change with starch (maize ugali flour used as the test substrate — connecting to the food-test skills from Strand 1.3). Students research (digital or print media) the properties of fluorine, bromine and iodine and compare trends down Group 17. Balanced equations for Cl_2 + H_2O and Cl_2 + NaOH are written. Students update DQB: "We can now explain bleaching — but what about elements that do NOTHING at all?" Lesson 5 — OBSERVE & EXPLAIN: Noble Gases — Stability and Inertness Students revisit the anchoring phenomenon: the argon welding shield and the helium balloon. Teacher explains full outer electron shells and the link to noble gas inertness (connecting back to Sub-Strand 2.1 electron configuration). Students research (digital devices / print media) the uses of noble gases: neon in road signage along Mombasa Road, argon in jua kali welding, helium in weather balloons used by Kenya Meteorological Department, krypton in photographic flash lamps. Groups create a mini-chart linking each noble gas to its use and the property that makes it suitable. Students update DQB: "Noble gases are stable — so how are the shiny, heavy transition metals different from all the families we've studied?" Lesson 6 — OBSERVE & INVESTIGATE: Transition Metals — Properties and Uses Students examine physical samples (or photographs) of copper wire, iron nails, zinc sheets (mabati), and lead weights and record physical properties (colour, lustre, hardness, conductivity). Teacher-guided discussion covers: variable oxidation states, catalytic behaviour, coloured compounds (copper sulphate solution — blue; iron(III) chloride — yellow-brown). Students connect to Kenyan contexts: copper in Kenyan electrical infrastructure, iron in jembes and SGR rails, zinc in roofing (mabati) across rural Kenya, lead in old Nairobi water pipes being replaced. Groups research (digital/print) additional transition metal uses and prepare a summary table. Students observe safety — handling dilute solutions with gloves and goggles, using first-aid kits if needed. DQB is updated with explanations for transition metal properties. Lesson 7 — EXPLAIN & MODEL: Connecting Properties to Uses Across All Chemical Families Students return to the full anchoring phenomenon and all four supporting phenomena. Working in groups, each group is assigned one chemical family and prepares a structured explanation poster (or digital slide): What are the family's key physical and chemical properties? What electron arrangement explains these properties? What are 3 Kenyan uses of elements from this family? Groups present to the class (peer teaching). Class collectively revises the element-sorting activity from Lesson 1 — students now correctly classify all 20 elements into families with justifications. Teacher facilitates a whole-class discussion on the driving question: "How are elements suited to their use in day-to-day life?" DQB is updated to reflect consolidated understanding. Lesson 8 — MODEL BUILDING, DQB CONSOLIDATION & ASSESSMENT Students individually (or in pairs) complete a cumulative model: a fully annotated chemical families reference chart that maps each family → key properties → electron arrangement pattern → 3 Kenyan uses → one safety/environmental consideration. Class conducts a final DQB review — circling questions now answered and flagging any remaining wonderings (linking forward to Sub-Strand 2.3 Chemical Bonding). Formative assessment: students write a short structured explanation responding to the driving question, using evidence from at least three investigations conducted during the unit. Teacher provides feedback and previews how chemical family properties determine the types of bonds elements form — the bridge to the next sub-strand.
Supporting Phenomena	– Supporting Phenomenon 1 — "Sodium vs. Magnesium in Water": A teacher demonstration (or video clip) shows a small piece

	of sodium metal dropped into a trough of water fizzing violently, skidding across the surface and occasionally igniting — while a piece of magnesium ribbon placed in the same trough barely reacts, producing only tiny bubbles over several minutes. Students are puzzled: both are shiny metals, yet their reactivity is dramatically different. This supports learning about alkali metals vs. alkaline earth metals and trends in reactivity within and between families. – Supporting Phenomenon 2 — "The Colour-Changing Flower Experiment": A fresh red hibiscus (commonly grown in Kenyan school compounds) placed in a jar with a few drops of chlorine water (or exposed to chlorine gas from dilute HCl + $KMnO_4$) loses its colour within minutes, turning white. Students are puzzled: how can an invisible gas destroy colour? This supports learning about the oxidising/bleaching properties of halogens and connects to the use of chlorine in Nairobi's water treatment plants. – Supporting Phenomenon 3 — "The Rusting Jembe vs. the Shiny Copper Wire": Students observe that an iron jembe (hoe) left in the Rift Valley red soil rusts quickly, while the copper wiring inside a broken radio stays shiny for years. Yet both are transition metals. This supports learning about the varied physical and chemical properties of transition metals and their differential uses in Kenyan agriculture, construction and electronics. – Supporting Phenomenon 4 — "Noble Gas Balloons vs. Hydrogen Balloons": Students recall (or watch a short video of) the 1937 Hindenburg disaster compared with the safe use of helium balloons at Kenyan school celebrations. Both gases are lighter than air, but one is catastrophically reactive and the other completely inert. This supports learning about the unique stability of noble gases and explains their safe industrial and commercial applications.

LESSON 1 (40 minutes): Family Resemblances: Launching the Chemical Families Phenomenon	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To launch the anchoring phenomenon by engaging students in noticing and wondering about the dramatically different behaviours of elements encountered in everyday Kenyan life, and to establish a Driving Question Board that will guide the entire unit of learning.
Knowledge	Students will recall prior knowledge of elements and electron arrangements from Sub-Strand 2.1 and begin to articulate that elements can be grouped into families based on patterns in their properties.
Skills	Students will practise observation, prediction, group discussion, card-sorting classification, and recording of questions and initial ideas on the Driving Question Board.
Attitudes	Students will demonstrate curiosity and open-mindedness when confronted with puzzling phenomena, and show respect for peers by listening actively during group and whole-class discussions.
Key Inquiry Question	How are elements suited to their use in day-to-day life?
Purpose in Storyline	This is the PREDICT opening lesson. Its role is to create productive puzzlement — students encounter the anchoring phenomenon before any instruction is given, surface prior knowledge through card-sorting, and generate the questions that the remaining seven lessons will answer. No answers are given today; the lesson ends with unresolved wondering.
Safety Notes	

B. LESSON OVERVIEW

Students are presented with four striking anchoring images or real objects — a sodium street lamp (representing Thika Road), a jua kali argon-welding shield, a calcium hypochlorite water treatment tablet fizzing in a cup of water, and a helium party balloon — and are asked to notice, wonder, and predict before any teaching occurs. Through structured group discussion and a card-sorting activity using the first 20 elements, students activate prior knowledge of electron arrangements from Sub-Strand 2.1 and begin intuitively grouping elements by perceived similarity. The lesson closes with the class co-constructing a Driving Question Board populated entirely by student-generated questions. The teacher introduces the term chemical family as a label for the puzzle, not as an answer.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase 📹 **VIDEO:** Video: Atomic Interactions Source: PhET Interactive Simulations (English) 🔍 Search ARES	Students enter the room to find four labelled stations or a single display table at the front containing: (1) a large photograph or printed image of the orange-glowing Thika Road sodium street lamps at night, (2) a photograph or short video clip (played on a phone	Teacher opens with NO explanation — purely a gesture toward the display: 'Before I say a single word about what these things are, just look. What do you notice? What puzzles you? Take 30 seconds of silent looking first.' [WAIT TIME: 30 seconds of	Notice-Wonder-Predict routine anchors thinking in concrete observation before abstraction. Deliberate silence before group talk ensures every student forms an individual impression before being influenced by peers. The four objects are chosen to	Teacher scans sticky-note predictions during circulation to gauge whether students are connecting the objects to prior knowledge of elements and the periodic table from Sub-Strand 2.1. Look for: Do students mention the word element or atom? Do any

for similar videos 📖 **HTML:** Chemical Bonding (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	or laptop) of a jua kali welder in Kamukunji with a bright blue-white flame and a small argon gas cylinder visible, (3) a clear plastic cup in which a half-tablet of calcium hypochlorite has just been dropped into water — it fizzes gently and gives off a faint sharp smell (the cup is sealed with cling film so students observe through the film without inhaling fumes), and (4) an inflated helium party balloon tied to the display table. Students work in groups of four or five. Each group receives a NOTICE-WONDER recording sheet with two columns: What do I notice? and What do I wonder? Groups spend 6 minutes silently and then collectively recording at least 4 notices and 4 wonders. Groups are then asked to attempt one prediction: What do these four substances have in common, if anything? Each group writes their prediction on a sticky note and holds it ready.	complete silence — teacher steps back and does not speak or gesture.] Teacher then says: 'Now talk in your groups. You have 6 minutes. Fill in your notice and wonder sheet. Try to write at least four things in each column. Do not worry about being right — there are no wrong observations today.' [WAIT TIME: Allow full 6 minutes of group talk — teacher circulates without answering questions, responding only with: 'That is a great wondering — write it down.' or 'What makes you say that? Tell your group.'] After 6 minutes teacher asks: 'Each group, write one prediction on your sticky note: What do these four substances have in common? Stick it on the board when you are ready.' Teacher collects sticky notes on one section of the board without commenting on correctness.	maximise cognitive dissonance: an orange glow, an invisible gas doing physical work, a fizzing white powder, and a harmless floating balloon — all are elements yet they appear utterly unrelated.	students mention electron arrangement? Are predictions purely observational (colour, smell) or do any reach toward structure? This informs how much scaffolding to provide in the card-sort that follows.
Observe Phase 📹 **VIDEO:** Video: Atomic Interactions Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** Chemical Bonding (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Teacher reveals the identity of each substance — sodium (in the lamp), argon (in the welding shield), chlorine compound (in the water tablet), and helium (in the balloon) — and writes the element names and symbols on the board. Students are then given a set of 20 element cards (pre-printed, one set per group). Each card shows: element name, symbol, atomic number, and one brief real-world Kenyan use (for example, Iron — Fe — 26 — used in jembes and SGR rails; Neon — Ne — 10 — used in road signs along Mombasa Road; Sodium — Na — 11 — used in Thika Road street lamps). Students are told: 'Without using any notes or textbook, sort these	After revealing element identities, teacher says: 'Notice that these four very different substances — a glowing light, a shielding gas, a fizzing tablet chemical, and a floating balloon gas — are all elements on the periodic table. That is our puzzle for this entire unit.' Teacher then distributes element card sets: 'Your job now is to be a scientist who has never been told the rules. Sort these cards into families — groups that seem to belong together. Use any pattern you notice. You have 7 minutes. Go.' [WAIT TIME: Allow 7 full minutes — circulate and ask probing questions only: 'What is your reason for putting those two together?' or 'Is there anything	The card-sort is an elicitation tool, not an instruction tool. By sorting without rules, students externalise their mental models of element similarity, making prior conceptions visible to the teacher and to themselves. The Kenyan use printed on each card connects abstract symbols to tangible local context, making the sorting personally meaningful. Anchoring four of the cards to the four phenomena the students just observed creates a direct sensory-to-symbol link.	Teacher photographs or mentally notes each group's card-sort arrangement. Key diagnostic questions: Are students using atomic number sequence (Sub-Strand 2.1 rote knowledge) or are they using property-based logic? Do any groups spontaneously create a group for gases that do nothing — anticipating noble gases? Do any students place sodium and potassium together — showing they recall Group 1 from the periodic table? These data points guide differentiation in Lessons 2 through 4.

	20 cards into groups that you think belong together. You can make as many groups as you like and call them whatever you like. You have 7 minutes.' Groups sort the cards and label each group with a group name they invent themselves (for example, shiny solids, gas that does nothing, reactive, glowing).	surprising about where sodium ended up?' Do NOT confirm or correct groupings.] After 7 minutes: 'Freeze. Do not change your groupings yet. We will come back to these cards at the end of the unit — and you will see how your scientific thinking has grown.'		
Explain Phase 📹 **VIDEO:** Video: Atomic Interactions Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** Chemical Bonding (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Groups share their card-sort groupings briefly — each group names their groups aloud and gives one reason for one grouping choice. Teacher uses a class-facing large version of the 20 element cards (drawn on the board or displayed on a screen) to show the range of student groupings without declaring any correct. Teacher then introduces — for the first time in this unit — the term chemical family: 'Scientists have noticed these same patterns for over 150 years. They gave the groups a name: chemical families. A chemical family is a group of elements that share similar properties because of their electron arrangement.' Teacher writes the definition on the board. Teacher asks students to look at their element cards again and circle any element whose electron arrangement they can recall from Sub-Strand 2.1. Teacher then connects explicitly: 'Remember in Sub-Strand 2.1 we described electron arrangement as the number of electrons in each shell. Today we are starting to ask: could the number of electrons in the outermost shell be what puts elements into the same family?'	Teacher facilitates share-out briskly — one sentence per group: 'Group one, name your groups and give us one reason — 20 seconds.' Repeat for all groups. Then teacher says: 'Nobody is wrong yet — because we are all still predicting. That changes over the next seven lessons.' Teacher introduces the definition: 'Write this down — chemical family: a group of elements with similar properties due to similar electron arrangements in their outer shells.' [WAIT TIME: 20 seconds for students to write.] Teacher then asks: 'Without looking anything up, raise your hand if you can tell me how many outer electrons sodium has.' [WAIT TIME: 10 seconds.] If students answer correctly (one outer electron), teacher says: 'Excellent — hold that thought. We will use it next lesson to explain why sodium makes the street lamp glow orange AND why it explodes in water.' If no student answers, teacher says: 'That is going to be our starting point in Lesson 2 — and it will explain the street lamp and the explosion. For now, the question is on the board.'	The Explain phase in Lesson 1 is intentionally minimal — it introduces one vocabulary term and one connecting question, not a full explanation. This is consistent with the storyline purpose: productive puzzlement, not premature closure. The connection to Sub-Strand 2.1 electron arrangement gives students a conceptual hook without removing the need to investigate further. The teacher deliberately leaves the sodium question unanswered if students cannot recall it, modelling that unanswered questions are the engine of science.	Listen for the quality of reasoning in the one-sentence share-outs. Are students using the word electron or shell spontaneously? Do any students connect the chemical family idea to the periodic table groups they encountered in Sub-Strand 2.1? This determines how much Sub-Strand 2.1 review to weave into the opening of Lesson 2.
Driving Question Board (DQB)	Each student receives three sticky notes of different colours: pink for questions I have, yellow for things	Teacher says: 'This board belongs to you — not to me. Every time we do an investigation and learn	The three-colour sticky-note protocol externalises three distinct cognitive states — curiosity (pink),	The DQB is a formative assessment artefact. Teacher analyses the distribution of pink,

Creation 📹 **VIDEO:** [Video: Atomic Interactions] Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** [Chemical Bonding (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	I think I know, and blue for things that surprised me. Students spend 3 minutes individually writing one item per sticky note — they may write more than one of any colour. Students then bring their sticky notes to a dedicated section of the classroom wall or board labelled OUR DRIVING QUESTION BOARD — Chemical Families Unit. The teacher reads out the driving question already written in large text at the top of the board: 'Why do some elements explode in water, some glow silently in glass tubes, and others rust on a Kenyan farm — and how do their family properties explain the materials that build, light up, and sustain our daily lives?' Students place their sticky notes and the teacher does a quick live sort, clustering similar questions together and labelling emerging clusters aloud: 'I see several questions about why some elements react and others do not — that goes here. I see questions about what is inside the atom that causes this — that goes here. I see questions about specific uses — that goes here.' Teacher leaves all clusters open — nothing is labelled as answered.	something new, we will come back and update it. By the end of Lesson 8, I want every pink question to have an answer — written by you, not by me.' Teacher reads the driving question slowly and deliberately: 'Listen to this question: Why do some elements explode in water, some glow silently in glass tubes, and others rust on a Kenyan farm? We just saw all of those things in our four phenomena today. This question is now your scientific mission.' [WAIT TIME: Allow 3 full minutes of silent individual sticky-note writing — no talking.] During the clustering, teacher narrates thinking aloud: 'This student asks why sodium makes an orange colour — brilliant question. I am putting that in the cluster about properties and uses. We will answer that in Lesson 2.' Teacher ends with: 'Notice that the board is full of questions and almost empty of answers. That is exactly how science starts.'	prior knowledge (yellow), and surprise (blue) — giving the teacher diagnostic data on all three simultaneously. The public, shared nature of the DQB creates collective intellectual ownership of the unit questions. By clustering questions live in front of students, the teacher models scientific categorisation and shows students that their individual wonderings connect to larger patterns — mirroring the concept of chemical families itself.	yellow, and blue notes after class. Key indicators: How many students placed yellow notes showing prior knowledge of electron arrangement? How many pink questions are about the specific phenomena observed today versus generic chemistry questions? Are any blue surprise notes about the Kenyan contexts specifically — for example, surprise that argon is used in jua kali welding — indicating that the local context is doing cognitive work? This analysis shapes the opening review strategy for Lesson 2.
Model Building Phase 📹 **VIDEO:** [Video: Atomic Interactions] Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:**	Students return to their element card-sort. Each group is given one additional blank card and asked to write a group label for the four anchoring phenomenon elements: sodium, argon, chlorine (from calcium hypochlorite), and helium. They must write: (1) the group name they are giving this collection, (2) one property all four share, and (3) one property that makes each one different from the others. This is the first draft of their	Teacher says: 'Your card-sort grouping from earlier — that is your first scientific model. It is not wrong. It is your current best thinking. Scientists revise their models as they gather more evidence. We are going to do the same over eight lessons.' Teacher distributes the blank card: 'Your last task today: make one new card for the four substances from our phenomenon. What would you call that group? What do they	The blank-card model-building task makes the transition from observation to representation — a key science practice. By explicitly labelling it Draft 1 and promising a final-draft comparison in Lesson 7, the teacher creates a longitudinal self-assessment structure that makes growth visible and motivating. The Science Journal entry is a low-stakes individual writing task that consolidates the group thinking into a personal	Collect or photograph the blank cards before students leave. The quality of the one shared property identified for all four elements is the key indicator: students who write all four are elements or all four are on the periodic table are at a surface level; students who attempt something like all four behave very differently from what I expected are demonstrating productive puzzlement; students who write something connecting to

Chemical Bonding (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	mental model — explicitly labelled as a first draft. The teacher collects or photographs each group's completed blank card. Students are told: 'At the end of Lesson 7, you will do this exact activity again — and you will see how your model has changed. That change IS your learning.' Students are given a Student Science Journal page titled My Chemical Families Model — Draft 1 and asked to sketch or write their current best thinking about what a chemical family is and why the four phenomenon elements are puzzling. This is completed in the last 2 minutes and kept in their journals for the entire unit.	share? What makes each one different? You have 3 minutes.' [WAIT TIME: Allow 3 full minutes.] Teacher then says: 'Open your Science Journal to the page labelled My Chemical Families Model — Draft 1. In 2 minutes, write or sketch your current best answer to this question: What is a chemical family and why are these four substances puzzling?' [WAIT TIME: Allow 2 minutes of silent writing.] Teacher closes: 'These journals stay with you for this unit. Do not lose this page. Lesson 2, we start with sodium — and I promise you, it will be the most dramatic chemistry demonstration you have seen in this classroom.'	record, ensuring that every student — not just the most vocal group members — has processed the lesson content. The teacher preview of Lesson 2 as dramatic maintains motivational momentum and connects the DQB questions to the next investigation.	electrons or outer shells are demonstrating strong Sub-Strand 2.1 recall that can be leveraged in Lesson 2. Science Journal Draft 1 entries are reviewed by the teacher before Lesson 2 to identify persistent misconceptions — particularly any belief that noble gases are not real elements or that reactive elements are dangerous and therefore less useful.

D. TEACHER REFLECTION

After the lesson, ask yourself: (1) Did every group engage with the card-sort, or were some groups dominated by one student — if so, assign roles (card reader, recorder, timekeeper, reporter) more explicitly in Lesson 2. (2) Were students genuinely puzzled by the phenomenon or did the familiarity of the objects (street lamps, balloons, welding) reduce the cognitive surprise — if so, emphasise in Lesson 2 the moment of sodium meeting water as a direct challenge to their intuitions. (3) Did the DQB generate at least 15 distinct questions from the class — if fewer than 15, the class may need more structured prompting in future lessons using sentence starters such as I wonder why... or I am confused about... (4) Did any student connect the chemical family idea to the periodic table groups from Sub-Strand 2.1 — if yes, this student can serve as a peer explainer in Lesson 2. (5) Were the Kenyan contexts (Thika Road, Kamukunji, Nairobi Water) resonant for students, or did some students indicate unfamiliarity — if unfamiliar, begin Lesson 2 with a brief 2-minute image gallery of these locations to build shared reference.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Students observed four anchoring phenomenon objects: a sodium street lamp photograph from Thika Road, a jua kali argon-welding image from Kamukunji, a calcium hypochlorite tablet fizzing in water, and a helium party balloon. Students observed that these four substances look completely different, behave completely differently, and are used for completely different purposes — yet all are elements.
What did I learn?	Students learned the term chemical family as a label for a group of elements with similar properties due to similar electron arrangements in their outer shells. Students recalled from Sub-Strand 2.1 that elements have electrons arranged in shells and that the outer shell is significant. Students learned that the class has a Driving Question Board that they will update over eight lessons as they gather evidence.
How does this explain the phenomenon?	Students cannot yet fully explain why the four phenomenon elements behave differently — this is intentional. Students have articulated their first-draft model: an initial grouping of 20 elements based on perceived similarity, and an initial characterisation of the four phenomenon elements. The gap between what students observed and what they can explain is the productive tension that

	drives the remaining seven lessons.

LESSON 2 (80 minutes (2 × 40-minute lessons)): Sorting the Periodic Table Into Families: Who Belongs Together?	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To introduce the concept of chemical families by having learners sort the first 20 elements into groups based on patterns in electron arrangement, connecting family membership to the dramatically different behaviours observed in the anchoring phenomenon.
Knowledge	Learners identify the five main chemical families in the periodic table — alkali metals, alkaline earth metals, halogens, noble gases, and transition metals — and classify the first 20 elements into their correct families based on electron configuration patterns from Sub-Strand 2.1.
Skills	Learners sort and group element cards using observable patterns (electron arrangement, valence electrons), construct an initial sorting rationale, record evidence in a science notebook, and communicate reasoning to peers using scientific vocabulary.
Attitudes	Learners appreciate that the seemingly chaotic differences between sodium (street lamp), argon (welding shield), chlorine compounds (water tablet), and helium (balloon) are actually explained by predictable family patterns, fostering confidence that chemistry is orderly and discoverable.
Key Inquiry Question	How are elements suited to their use in day-to-day life?
Purpose in Storyline	Lesson 1 established the puzzle: four very different substances — sodium, argon, chlorine compounds, helium — are all made of elements, yet behave completely differently. Lesson 2 is the first evidence-gathering step: learners discover that elements can be sorted into families sharing electron-arrangement patterns, giving them the conceptual tool they need to begin explaining the phenomenon. This lesson activates and extends prior knowledge from Sub-Strand 2.1 (Periodic Table) and sets up Lessons 3–8 in which each family's properties are investigated in depth.
Safety Notes	This lesson is primarily card-sorting and discussion — no chemicals are handled. If element sample images or sealed demonstration vials are displayed, learners must not open any sealed containers. Teacher must remind learners that sodium metal and chlorine gas (referenced in images/videos) are hazardous; they are observed only through photographs and video in this lesson. First-aid kit must be present in the room as per school protocol.

B. LESSON OVERVIEW

Learners revisit the four anchoring images from Lesson 1 (Thika Road sodium lamp, jua kali argon-welding, calcium hypochlorite fizzing in water, helium party balloon) and add two new objects: a jembe (iron hoe) and a copper electrical wire. Working in groups of four, learners receive a set of 20 element cards (each card shows element name, symbol, atomic number, and electron arrangement) and attempt to sort them into families without being told the rules — using only patterns they can spot. Groups then share their sorting logic in a gallery walk. The teacher introduces the formal family names (alkali metals, alkaline earth metals, halogens, noble gases, transition metals) and learners re-sort their cards using the new vocabulary. The lesson closes with learners updating the class Driving Question Board (DQB) and recording a revised model of why the anchoring phenomenon substances behave differently.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy

1 — Predict Phase (15 minutes) 📹 **VIDEO:** [Naming ions and ionic compounds](#) Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Chemical Bonding (Flexbook)](#) Source: CK-12 🔍 Search ARES for similar readings	Learners examine six objects/images displayed at the front of the class: (1) a photograph of Thika Road sodium street lamps glowing orange at night, (2) a photograph of a jua kali welder in Kamukunji using an argon-shielded torch, (3) a cup of water with a calcium hypochlorite tablet fizzing in it, (4) a helium party balloon, (5) a jembe (iron hoe) with a rusted blade, and (6) a copper electrical wire stripped of its insulation. Each group of four receives a 'Prediction Slip' with three questions: (a) Which of these six objects do you think are made from elements that 'belong together'? Draw lines connecting the ones you think are in the same family. (b) What pattern or clue are you using to decide? (c) What do you think the word 'chemical family' means? Learners write individual predictions silently for 3 minutes, then discuss within their group for 4 minutes and agree on a group prediction to post on the class Prediction Wall.	Display all six objects/images clearly — use printed A3 photographs if a projector is unavailable. Distribute Prediction Slips. Say: 'Before we learn anything new today, I want you to use only what you already know from Sub-Strand 2.1 to make a prediction. There is no wrong answer at this stage — your job is to think.' Allow 3 minutes of SILENT individual writing (no discussion). After 3 minutes, say: 'Now share your thinking with your group. You have 4 minutes to agree on one group prediction.' Circulate and listen — do NOT correct predictions. After groups post their Prediction Slips, cold-call 3 groups (choose one high-confidence group, one uncertain group, one group with a surprising idea) and ask: 'Tell us ONE clue you used to group these elements together.' Allow 10 seconds of WAIT TIME after each question before accepting a response. Record key prediction words on the board (e.g., 'shiny,' 'gas,' 'reacts,' 'doesn't react') — these will be revisited in the Explain Phase.	Elicit-then-Compare Prediction: By committing predictions to paper before instruction, learners activate prior knowledge from Sub-Strand 2.1 (electron arrangement, atomic number) and create a cognitive 'hook' — a prediction they will be motivated to test. Posting predictions on the wall makes thinking public, enabling the teacher to identify misconceptions (e.g., 'all gases are in the same family') that will be deliberately addressed in Phase 3.	Teacher scans Prediction Slips during circulation. Look for two specific indicators: (1) Does the learner reference electron arrangement or valence electrons as a grouping clue — evidence of transfer from Sub-Strand 2.1? (2) Does the learner group sodium with potassium (both alkali metals) or helium with argon (both noble gases) — evidence of intuitive pattern recognition? Flag any group that groups elements only by physical state (all solids together, all gases together) for targeted questioning in Phase 3.
2 — Observe Phase (20 minutes) 📹 **VIDEO:** [Naming ions and ionic compounds](#) Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Chemical Bonding (Flexbook)](#)	Each group receives a set of 20 laminated Element Cards (H, He, Li, Be, B, C, N, O, F, Ne, Na, Mg, Al, Si, P, S, Cl, Ar, K, Ca). Each card displays: element name, symbol, atomic number, mass number, and electron arrangement written in the format used in Sub-Strand 2.1 (e.g., Na: 2,8,1). Groups also receive a blank 'Sorting Mat' — a large sheet of paper divided into between 3 and 6 columns (the number of columns is deliberately left blank for learners to decide). Task: Sort all	Distribute Element Card sets and Sorting Mats. Say: 'Your only tool is the electron arrangement written on each card. Look for patterns. You do NOT need to know the family names yet — just find the pattern and name the group yourself.' Circulate actively during the 10-minute sort. At the 5-minute mark, if a group is stuck, use a Socratic prompt — do NOT give answers: 'How many electrons does lithium have in its outermost shell? How about sodium? How about potassium? What do you	Pattern-Finding Card Sort (NGSS Science and Engineering Practice 8 — Obtaining, Evaluating, and Communicating Information): The card sort forces learners to interrogate the electron arrangement data on each card rather than receive a pre-organised table. By constructing their own grouping rules, learners develop the intellectual ownership of the classification concept. The Gallery Walk adds a peer-review layer, exposing learners to alternative sorting logics and	Observable indicator 1: All groups should correctly identify at least one 'noble gas' column (He, Ne, Ar all have full outer shells: 2, 2,8, 2,8,8) — if a group has split these three elements across different columns, the teacher must intervene with a targeted question during the Gallery Walk. Observable indicator 2: Groups should notice that Li, Na, K all have 1 valence electron and place them together. If this column is absent, note the group for re-teaching in Phase 3. Collect one

Source: CK-12 🔍 Search ARES for similar readings	20 cards into groups. Write a 'family rule' for each group — one sentence explaining what all members of that group have in common. Groups have 10 minutes. After sorting, groups conduct a 4-minute Gallery Walk: they leave their Sorting Mat on the desk, rotate clockwise to the next group's mat, and use sticky notes to write one 'I notice…' and one 'I wonder…' comment on the mat they are visiting.	notice?' (WAIT TIME: 12 seconds after each question.) After the Gallery Walk, cold-call 2 groups and ask: 'What was the most surprising thing you noticed on another group's mat?' (WAIT TIME: 10 seconds.) Then ask the whole class: 'Which column on your mat do you think contains the most reactive elements — and what in the electron arrangement makes you think that?' Accept 3 responses, record on board without confirming or denying.	prompting metacognitive 'I wonder…' questions that feed the DQB in Phase 4.	Sorting Mat from each group to review after the lesson.
3 — Explain Phase (25 minutes) 📹 **VIDEO:** Naming ions and ionic compounds Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Chemical Bonding (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	The teacher reveals the formal family names and connects them to the learners' own sorted columns using a structured 'Name Your Column' activity. Each group receives a 'Chemical Families Reference Card' listing the five families, their group numbers, valence electron counts, and one Kenyan real-life connection (e.g., 'Group 1 — Alkali Metals: Na is in the Thika Road street lamp; K is in fertilisers used by maize farmers in the Rift Valley'). Learners re-label their Sorting Mat columns with the official family names. Then, as a whole class, learners work through four 'Phenomenon Connection Questions' in their science notebooks: (1) Sodium is in Group 1. It has 1 valence electron. What does this tell you about how easily it reacts? (2) Argon is in Group 0. Its outer shell is full. What does this predict about whether argon will react with the metal being welded? (3) Chlorine is in Group 7. It has 7 valence electrons. What does this suggest about its 'eagerness' to gain one more electron — and why does it react with water in the storage tank? (4) Helium is also in Group 0, like	Distribute Chemical Families Reference Cards. Say: 'Look at your sorted columns. Now I am going to give each column its real scientific name. Your job is to check — does the official family match the pattern YOU discovered?' Give groups 2 minutes to re-label, then ask the class: 'Which group found all five families on their own — raise your hand.' Acknowledge effort. Then move to the Phenomenon Connection Questions. Read Question 1 aloud and say: 'Write your answer silently first. You have 90 seconds.' (Enforce silence.) Then say: 'Turn to your partner. Share your answer. Do you agree?' (2 minutes.) Cold-call 3 learners for Question 1 using name sticks — ensure gender balance across cold-calls. Ask follow-up: 'Can you connect your answer back to the sodium street lamp on Thika Road?' (WAIT TIME: 12 seconds.) Repeat the same sequence for Questions 2, 3, and 4, spending approximately 3 minutes per question. After Question 4, write on the board: 'FAMILY RULE: Same number of valence electrons → Similar	Phenomenon-Anchored Concept Mapping (NGSS Science and Engineering Practice 6 — Constructing Explanations): Rather than teaching the family names abstractly, every family is immediately connected to one of the six anchoring objects. This 'anchor-to-concept' sequence ensures that the abstract concept of valence electrons is always grounded in observable, Kenyan-context phenomena. The Phenomenon Connection Questions scaffold learners from data (electron arrangement) to explanation (behaviour) to application (real-world use), building the chain of reasoning they will need throughout the unit.	Exit check for Question 4 (helium in balloon): A complete answer must contain two elements — (a) helium has a full outer electron shell (2 electrons, group 0) and (b) a full outer shell means it does not react / gain or lose electrons easily, making it safe and inert. If learners write only 'helium is a noble gas' without referencing the electron arrangement, mark as 'approaching expectations' and address in Lesson 3 opener. Target: ≥ 75% of learners give a complete two-part answer before the lesson ends.

	argon. Why is it safe in a party balloon? Learners write individual answers (5 minutes), then pair-share (3 minutes), then the class discusses (5 minutes). Teacher introduces the term 'valence electrons' as the key to family behaviour and writes the summary rule on the board: 'Elements in the same chemical family have the same number of valence electrons — and this is WHY they behave similarly.'	chemical behaviour.' Ask: 'In one sentence in your notebook, explain WHY argon is used as a welding shield gas in the jua kali workshop in Kamukunji.' Collect 4 notebooks at random to check responses before Lesson 3.		
4 — Driving Question Board (DQB) Update (10 minutes) 📹 **VIDEO:** [Naming ions and ionic compounds] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Chemical Bonding (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	The class DQB (created in Lesson 1) is displayed at the front. Each group reviews their Lesson 1 'wonderings' and does three things on sticky notes: (1) ANSWERED sticky (green): Write one question from Lesson 1 that they can now partially or fully answer — and write a one-sentence answer beneath it. (2) REFINED sticky (yellow): Write one question from Lesson 1 that has become more specific now that they know about chemical families (e.g., upgrade 'Why does sodium react with water?' to 'Does every alkali metal react with water the same way as sodium?'). (3) NEW WONDER sticky (orange): Write one brand-new question that arose from today's sorting activity. Groups post all three stickies on the DQB. Teacher reads 3–4 new orange 'wonder' stickies aloud to the class and says: 'These are the questions that will drive Lessons 3 through 8. Keep them in mind.'	Point to the DQB and say: 'In Lesson 1 you asked great questions. Today you gathered new evidence. Let's see how our thinking has grown.' Distribute three sticky-note colours to each group. Allow 4 minutes for groups to write all three stickies. Circulate and prompt any group that is stuck on the 'ANSWERED' sticky: 'Look at your Prediction Slip from the start of today's lesson — what did you predict that you can now explain better?' After posting, read 3 orange 'NEW WONDER' stickies aloud (choose ones that preview upcoming lessons — e.g., ones about reactivity, uses of metals, or noble gas applications). Say: 'These are excellent science questions. A scientist in Kenya who works with the Kenya Atomic Energy Authority or KEBS would ask exactly these kinds of questions about materials.' Write the top 2 new wonder questions on a 'Priority Questions' section of the DQB for Lesson 3.	Driving Question Board Curation (NGSS Science and Engineering Practice 1 — Asking Questions and Defining Problems): The three-colour sticky-note protocol makes the growth of understanding visible and cumulative. 'ANSWERED' stickies give learners a sense of progress and consolidate new knowledge. 'REFINED' stickies develop the scientific habit of making questions more precise as evidence accumulates. 'NEW WONDER' stickies sustain curiosity and provide the teacher with real-time data on which aspects of chemical families are most puzzling to the class, allowing adaptive planning for Lessons 3–8.	Teacher photographs the DQB after the lesson. Count: (a) Number of green 'ANSWERED' stickies that correctly reference valence electrons or family membership — target ≥ 60% of groups. (b) Number of yellow 'REFINED' stickies that successfully narrow a broad question into a family-specific testable question — this indicates readiness for the deeper investigation in Lesson 3. (c) Flag any group that cannot produce a green 'ANSWERED' sticky — they may need a brief recap at the start of Lesson 3.
5 — Model Building Phase (10 minutes) 📹 **VIDEO:**	Each learner opens their science notebook to their Lesson 1 Initial Model — a drawing or diagram they made in Lesson 1 to try to explain why sodium, argon,	Say: 'Take out your Lesson 1 model. Using a different coloured pen, you are going to upgrade it with today's evidence. You have 6 minutes.' Circulate and look for	Cumulative Model Revision (NGSS Science and Engineering Practice 2 — Developing and Using Models): Using a different-coloured pen for each lesson's	Spot-check 5 notebooks at random. Evaluate the revised model against three criteria: (1) All six anchoring objects are labelled with the correct family name ✓/✗.

Naming ions and ionic compounds Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Chemical Bonding (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	chlorine, and helium behave so differently. Using a coloured pen (different from Lesson 1), learners revise their model to include: (a) Labels showing which chemical family each of the six anchoring objects belongs to. (b) An arrow or annotation linking each family label to the number of valence electrons. (c) A one-sentence 'Model Claim' at the bottom: 'I now think the different behaviours are caused by ___ because ___.' Learners share their revised model with their partner using a 30-second 'model talk' — one partner presents, the other asks one clarifying question.	models that still lack the valence electron link — prompt: 'Your model shows sodium in Group 1. What does being in Group 1 tell us about sodium's outer electrons? Add that to your model.' After 6 minutes, say: 'Do a 30-second model talk with your partner. One person presents, the other asks: What does your model NOT yet explain?' After 30 seconds, cold-call 2 learners to share their 'Model Claim' sentence. Write both claims on the board and ask the class: 'What is similar about these two claims? What is still missing — what can our model NOT yet explain?' (WAIT TIME: 12 seconds.) Say: 'In Lesson 3, we will investigate the actual physical and chemical properties of alkali metals so we can make our model even more accurate. Keep your notebook safe.'	revision creates a visible record of conceptual growth — learners can literally see their understanding deepen across the unit. The 'Model Claim' sentence forces learners to commit to an explanation at the current level of evidence, making it easy to identify and celebrate revision in later lessons. The 'What does your model NOT yet explain?' prompt sustains productive intellectual humility and links directly to the driving question.	(2) Valence electron count is linked to family membership ✓/✗. (3) The 'Model Claim' sentence connects electron arrangement to behaviour (not just to family name) ✓/✗. Any learner scoring 0/3 or 1/3 should be identified for a 5-minute check-in at the start of Lesson 3.

D. TEACHER REFLECTION

After the lesson, ask yourself: (1) During the card sort, did at least two-thirds of groups correctly identify the noble gas family (He, Ne, Ar) independently? If not, the electron arrangement patterns from Sub-Strand 2.1 may not be sufficiently consolidated — consider beginning Lesson 3 with a 5-minute electron arrangement recap using maize kernels to represent electrons in shells. (2) Were learners able to connect 'full outer shell' to 'inert/unreactive' when explaining argon in the Kamukunji welding context? This conceptual link is foundational for the rest of the unit — if fewer than half the class made this connection in their notebook, introduce a simple analogy at the start of Lesson 3: 'A shopkeeper whose shelves are completely full does not need to buy anything — similarly, an atom with a full outer shell does not need to gain or lose electrons.' (3) Did the DQB orange 'wonder' stickies reveal questions about specific families (alkali metals, halogens) or more general questions about the periodic table? If learners are still asking general questions, the family-specific focus of Lessons 3–5 is well-timed. (4) Was there equitable participation across gender and across groups of different prior attainment during cold-call questioning? Use your name-stick records to check. (5) Note any Kenyan-context connections that excited particular learner interest (e.g., the Rift Valley maize farmer / potassium fertiliser link, or the Kamukunji jua kali welding context) — these can be amplified in future lessons to sustain engagement.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Learners sorted 20 element cards into family groups based on electron arrangement patterns and observed that elements sharing the same number of valence electrons could be placed in the same column. They observed that sodium (street lamp), potassium (Rift Valley fertiliser), and lithium all have 1 valence electron; that helium and argon (welding shield, party balloon) both have full outer shells; and that chlorine has 7 valence electrons — one short of a full shell.

What did I learn?	Learners learned that the periodic table is organised into chemical families — alkali metals (Group 1), alkaline earth metals (Group 2), halogens (Group 7), noble gases (Group 0), and transition metals — and that membership in a family is determined by the number of valence electrons. Elements in the same family have the same number of valence electrons and therefore behave similarly. The dramatically different behaviours of sodium (reactive), argon (inert), chlorine (reactive, disinfecting), and helium (inert) seen in the anchoring phenomenon can be explained by their different family memberships.
How does this explain the phenomenon?	The anchoring phenomenon (sodium lamp glowing orange, argon shielding a Kamukunji weld, chlorine compound fizzing in a water tank, helium floating inertly in a balloon) is explained by valence electron count: sodium's single valence electron makes it highly reactive and able to release energy as light in a discharge lamp; argon's and helium's full outer shells make them completely unreactive — perfect as protective or inert gases; and chlorine's 7 valence electrons make it 'eager' to gain one more, driving its strong reactivity with water and its disinfecting power. Same family = same valence electrons = similar behaviour.

LESSON 3 (40 minutes): Alkaline Earth Metals — Moderate Reactivity and the Chemistry Behind Kenyan Fertilisers	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	Students will investigate the reactivity of magnesium as a representative alkaline earth metal and compare Group 2 behaviour to Group 1, using electron arrangement to explain differences, then connect findings to real Kenyan agricultural contexts.
Knowledge	Students will know that alkaline earth metals have two valence electrons, that this makes them less reactive than alkali metals but still significantly reactive, that reactivity increases down Group 2, and that calcium and magnesium compounds are used in Kenyan fertilisers such as Calcium Ammonium Nitrate (CAN).
Skills	Students will conduct a supervised experiment recording observations systematically, write balanced chemical equations for magnesium reacting with dilute hydrochloric acid and with steam, and compare reactivity trends across Groups 1 and 2 using evidence.
Attitudes	Students will demonstrate patience and precision in experimental observation, show curiosity about how atomic structure determines real-world usefulness, and appreciate the relevance of chemistry to food security in Kenya.
Key Inquiry Question	How are elements suited to their use in day-to-day life? — specifically, how does having two outer electrons make magnesium and calcium useful in agriculture without being dangerously reactive like sodium?
Purpose in Storyline	In Lesson 2 students explained why alkali metals explode or fizz violently in water by pointing to their single valence electron. Lesson 3 shifts focus to Group 2 so students can gather comparative evidence: two valence electrons produce moderate, controllable reactivity. This evidence advances the DQB and builds the pattern students will need in Lessons 4 and 5 when they encounter halogens and noble gases at the opposite reactivity extremes.
Safety Notes	

B. LESSON OVERVIEW

Students open by revisiting the anchoring phenomenon — the Thika Road sodium lamp flaring and the argon welding shield sitting quietly — and are challenged to predict whether calcium, found in the fertiliser bags stacked at Rift Valley agrovet shops, behaves more like explosive sodium or inert argon. They then carry out a hands-on experiment with magnesium ribbon reacting with dilute HCl, observe a teacher demonstration of magnesium reacting with steam, record observations, write balanced equations, and construct an electron-arrangement explanation for why Group 2 is less reactive than Group 1 but still far from inert. The lesson closes with students updating their Driving Question Board and adding Group 2 to their element-sorting cards, preparing them for the halogen investigations in Lesson 4.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase 📹 **VIDEO:** Worked example: Finding the formula of an	Students (still seated in their groups from Lesson 2) are shown two images side by side on the board or printed as an A4 handout: Image A — the Thika Road	Teacher opens with: 'Before we touch any equipment today, I want your brains working first. Look at these two images. You have told me that sodium — Group 1 — has	Prediction T-chart on the board anchors prior knowledge (Group 1 electron arrangement) and creates cognitive tension that motivates the experiment. Connecting to the	Teacher reads predictions written in student books during the 90-second wait time. Key diagnostic: do students correctly invoke electron arrangement as their

ionic compound Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Alkaline Earth Metals (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	sodium street lamp glowing orange (anchoring phenomenon); Image B — a 50 kg bag of CAN fertiliser (Calcium Ammonium Nitrate) leaning against the wall of an agrovet shop in Nakuru. Teacher asks: 'Both sodium in that lamp and calcium in that fertiliser bag are metals in the first two columns of the periodic table. Based on what we discovered about alkali metals in Lesson 2, predict: will calcium react with water the way sodium does — violently? Or will it be much calmer? Write your prediction in one sentence and give one reason using electron arrangement.' Students write individually for 90 seconds, then share with a partner for 60 seconds. Two or three pairs share aloud. Teacher records key prediction phrases on the board in a T-chart labelled 'Predicting Group 2 Reactivity': left column 'Will be like sodium — violent' and right column 'Will be calmer — two electrons harder to lose'. Teacher does NOT confirm which side is correct, preserving productive uncertainty. Students also retrieve their element-sorting cards from Lesson 1 and place magnesium (Mg) and calcium (Ca) cards face-up on their desks as reference points for the investigation.	one outer electron and loses it easily to react violently with water. Now look at Group 2. Magnesium and calcium sit right next to Group 1 on the periodic table. What does having TWO outer electrons do to reactivity? Make your best prediction — there is no wrong answer right now, we are scientists hypothesising.' [WAIT TIME: 90 seconds silent individual writing — teacher circulates, reads predictions over shoulders, notes any misconceptions such as students thinking more electrons always means more reactivity.] Teacher then says: 'Share with your partner — do you agree or disagree? If you disagree, whose reasoning is stronger?' [WAIT TIME: 60 seconds partner discussion.] Teacher cold-calls two groups with contrasting predictions and records both on the T-chart without evaluating. Teacher closes the Predict phase with: 'We have two competing ideas on the board. Our experiment in the next few minutes will give us the evidence to decide. Scientists do not guess — they test. Let us test.'	CAN fertiliser bag — a familiar object from Rift Valley farming households — makes the chemistry personally relevant before a single reaction has been observed.	reasoning, or do they rely on surface features like 'calcium is heavier'? Students who invoke electron arrangement correctly are identified to be called on during the Explain phase. Misconceptions noted here are addressed explicitly in the Explain phase.
Observe Phase 📹 **VIDEO:** Worked example: Finding the formula of an ionic compound Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos	The class splits into two parallel activities running simultaneously for approximately 12 minutes. ACTIVITY A (Student experiment — all groups): Each group bench has a test-tube rack with 3 test tubes each containing 5 mL of dilute HCl (0.5 M), a small strip of magnesium ribbon (approximately 2 cm) per test tube, forceps, a wooden splint, a box of matches,	Teacher distributes equipment and says: 'Before you begin, read the procedure card completely — all four steps. Do not add anything to any test tube until every person in your group has their goggles on. I am watching.' [WAIT TIME: 30 seconds for groups to read and equip goggles.] Teacher circulates actively during the experiment, asking probing questions at each	Parallel activities maximise student time on task within the 40-minute lesson. The structured observation table (describe — infer — gas test) scaffolds scientific recording skills without doing the thinking for students. The teacher demonstration of the steam reaction creates a visual contrast that students will need in the Explain phase: magnesium reacts	Teacher circulates and checks observation tables for precision: are students writing qualitative descriptions ('vigorous effervescence, ribbon dissolved completely in 45 seconds') or vague entries ('it reacted')? Teacher prompts for more precise language where needed. The round-robin at the end of the Observe phase gives teacher a

📖 **HTML:** [Alkaline Earth Metals (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	and a safety goggles set per student. Students follow a printed procedure card: (1) Put on goggles and aprons. (2) Using forceps, drop one magnesium ribbon strip into test tube 1. (3) Observe immediately: record colour change, any gas production (describe the sound and rate of bubbling), any temperature change felt on the outside of the test tube. (4) After 30 seconds, bring a lit splint to the mouth of the test tube — record what happens. (5) Repeat with test tubes 2 and 3 for reproducibility. Students record observations in a structured table: Column headers — Observation (describe exactly what you see/hear/feel), Inference (what does this tell you about the reaction), Gas test result. ACTIVITY B (Teacher demonstration — students observe from benches): Teacher uses a Bunsen burner to heat water in a boiling tube to produce steam, then introduces a coiled magnesium ribbon strip into the steam stream using long tongs. Students observe from their seats (minimum 1 metre distance) and record separately: does magnesium react with steam? Describe what you see. Both activities are completed within 12 minutes. Groups then share observations verbally in a 2-minute round-robin: each group states ONE thing they observed that surprised them.	bench: 'What do you see happening to the ribbon itself — is it disappearing, staying the same, changing colour?' and 'You tested the gas with the splint — what does a squeaky pop tell you about which gas was produced?' Teacher does NOT provide answers — only directs attention. For the steam demonstration teacher narrates quietly: 'I am keeping this ribbon away from the flame itself — I want it to meet only the steam. Watch the ribbon carefully.' After the demonstration: 'Did magnesium react with steam the same way it reacted with cold dilute acid? Faster? Slower? Different products? Write it down.' [WAIT TIME: 60 seconds for student recording after the demonstration.] During the 2-minute round-robin teacher listens for the key observation — 'bubbles formed and the splint gave a squeaky pop indicating hydrogen gas' — to confirm the reaction occurred. If no group mentions the splint test result, teacher asks: 'Did anyone get a result from the lit splint test? What does that result tell us about the gas produced?'	with steam but much less readily than sodium reacts with cold water — direct comparative evidence for the reactivity difference between Groups 1 and 2.	whole-class snapshot of what was noticed. Red flag: if multiple groups report no bubbles, teacher checks acid concentration and ribbon cleanliness (oxide layer on Mg can slow reaction — teacher models scraping ribbon with sandpaper if needed).
Explain Phase 📹 **VIDEO:** [Worked example: Finding the formula of an ionic compound]	Students now work in groups to construct explanations for their observations, guided by three questions printed on the procedure card: (Q1) Write a word equation and then a balanced symbol	Teacher opens the Explain phase: 'You have observations — now we need equations. Equations are the language scientists use to record exactly what happened. Work on Q1 in your groups for 3 minutes.	The scaffolded equation construction (skeleton first, then balancing) prevents cognitive overload while ensuring students engage with the chemistry rather than just copying. The side-by-side	Teacher checks balanced equations in student books during the 3-minute group work window — specifically looking for correct products ($MgCl_2$ and H_2) and correct balancing (coefficient 2 in

Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Alkaline Earth Metals (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	equation for magnesium reacting with dilute hydrochloric acid. (Q2) Write a word equation for magnesium reacting with steam to form magnesium oxide and hydrogen gas. (Q3) Using electron arrangement diagrams (draw the shells for Mg: 2, 8, 2), explain why magnesium is less reactive than sodium (2, 8, 1) but more reactive than argon (2, 8, 8). Students attempt Q1 and Q2 independently for 3 minutes before teacher writes the skeleton equations on the board and asks groups to complete the balancing. Teacher then leads a whole-class discussion connecting electron arrangement to the reactivity comparison. Students update their notes with the following key statement (teacher dictates): 'Alkaline earth metals have two valence electrons. These two electrons are lost during chemical reactions to form 2+ ions. Losing two electrons requires more energy than losing one, so Group 2 metals are less reactive than Group 1 metals. Reactivity still increases down Group 2 as the outer electrons are held less tightly by the nucleus.' Teacher then introduces the Kenyan context: 'The calcium in CAN fertiliser — Calcium Ammonium Nitrate — used by maize farmers in the Rift Valley is a calcium compound. Why is it safe to store in sacks and spread on fields without exploding? Because calcium is reactive enough to release nutrients in soil water slowly, but not so reactive that it ignites on contact with moisture the way potassium would. The two outer electrons make Group 2 elements controllably reactive — useful, not	Do not look at your Lesson 2 notes yet — try from memory first.' [WAIT TIME: 3 minutes group work.] Teacher writes on board: 'Mg + HCl → _____ + _____' and asks: 'What were the products? Your observations of the gas and the solution after the reaction should tell you.' Cold-calls a group: 'Group 3 — what did you write for products?' After hearing the answer, teacher completes the skeleton and asks: 'Is this equation balanced? Count the atoms on each side.' [WAIT TIME: 45 seconds.] Teacher writes the balanced equation: $Mg + 2HCl \rightarrow MgCl_2 + H_2$ and says: 'This is the balanced equation. Copy it and circle the coefficient 2 in front of HCl — that two is the balancing step. Now Q2 — the steam reaction.' Teacher repeats the process for $Mg + H_2O(steam) \rightarrow MgO + H_2$. For Q3 teacher draws electron shells for Na (2,8,1) and Mg (2,8,2) side by side on the board and asks: 'What is the only difference between these two atoms in terms of outer electrons?' [WAIT TIME: 30 seconds.] 'If sodium loses its one outer electron easily, what does Mg have to do differently — and does that make it easier or harder?' [WAIT TIME: 45 seconds.] Teacher connects to fertiliser context using the quote above, then closes: 'Update your element cards now. In 30 seconds I want to see Mg and Ca cards with at least three pieces of information on them.'	electron shell diagrams of Na and Mg make the abstract electron-arrangement comparison concrete and visual. The CAN fertiliser context transforms the Explain phase from abstract chemistry into a meaningful answer to the key inquiry question: elements are suited to their uses because of their family properties.	front of HCl). Common error to watch for: students writing MgCl instead of $MgCl_2$, reflecting confusion about Mg forming a 2+ ion. Teacher addresses this by pointing back to the electron arrangement: 'Mg loses TWO electrons — so it needs TWO chloride ions to balance. The equation shows us the electron story.' Students annotating their element cards provides a quick visual check that key ideas have been captured in their own words.

	dangerous.' Students annotate their Mg and Ca element-sorting cards with: family name, number of valence electrons, reactivity comparison to Group 1, and one Kenyan use.			
Driving Question Board (DQB) Creation 📹 **VIDEO:** Worked example: Finding the formula of an ionic compound Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Alkaline Earth Metals (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Each group receives two sticky notes in different colours. On the FIRST sticky note (green — 'We now know') students write one statement completing: 'We can now explain why alkaline earth metals are less reactive than alkali metals because...' On the SECOND sticky note (yellow — 'We still wonder') students write one new question that the lesson has raised. Groups bring their sticky notes to the class DQB at the front of the room. Teacher reads all 'We now know' notes aloud quickly (30 seconds each) and places them in the 'Answered' column of the DQB. Teacher reads the 'We still wonder' notes and groups them visually — likely clusters will be around: 'What about elements that have even MORE outer electrons?', 'Why do some non-metals react completely differently from metals?', and 'What makes chlorine in the water-purification tablet so reactive if it is not even a metal?' Teacher points to the chlorine question cluster and says: 'Look — several groups are already wondering about chlorine and the other reactive non-metals. That is exactly where we are going in Lesson 4. Your curiosity is pointing us in the right direction.' Teacher also asks: 'Where does today's evidence push our answer to the big driving question — why do some elements explode in water while others rust on a Kenyan farm? Can we say	Teacher distributes sticky notes and says: 'Two notes — two minutes. The green one is for what we can now explain that we could not explain at the start of this lesson. The yellow one is for a question that this lesson has opened up — not answered — opened up. Scientists always end an experiment with MORE questions, not fewer. Go.' [WAIT TIME: 2 minutes silent individual writing before groups compare notes and choose one of each colour to post.] As groups approach the DQB teacher says: 'Place your green note in the Answered column, your yellow note in the New Questions column.' Teacher reads notes aloud without editorial comment, then clusters yellow notes physically: 'I notice three groups are asking about reactive non-metals. I notice two groups are asking about elements in the middle of the table. These are real scientific questions — write them in your books under the heading: Questions I am carrying into Lesson 4.' Teacher closes DQB update with: 'Our driving question is about elements that explode, glow, rust, and disinfect. Today we explained one more piece — Group 2 reactivity. Next lesson the mystery deepens — we meet the halogens, the reactive non-metals responsible for that bleaching smell in the water tank.'	The two-colour sticky-note protocol makes the DQB update a structured thinking routine rather than an open-ended free-for-all. Clustering yellow notes publicly models how scientists identify productive research directions. Explicitly naming the upcoming halogen lesson as the destination for students current questions creates narrative momentum across the unit storyline.	Teacher reads all green notes before posting — this is a rapid whole-class formative check. Green notes that are vague ('we learned about Group 2') are gently pushed: 'Can you add the word electrons to that sentence to make it more precise?' Green notes that correctly invoke electron arrangement ('because two outer electrons require more energy to remove than one') confirm understanding. Yellow notes that reveal persistent misconceptions (e.g., 'Is calcium the same as calcium in milk?') are noted and addressed in the next lesson introduction.

	something more precise now?' Students suggest refinements and teacher writes a revised class consensus statement on the DQB: 'Elements behave differently because of how many valence electrons they have — the number determines how easily electrons are lost or gained and therefore how reactive the element is.'			
Model Building Phase 📹 **VIDEO:** Worked example: Finding the formula of an ionic compound Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Alkaline Earth Metals (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	In the final 5 minutes students open their cumulative reference chart — the running model they began building in Lesson 1 and updated in Lesson 2. They add a new row for Group 2 Alkaline Earth Metals. The row has five columns: Family Name | Key Properties | Electron Arrangement Pattern | Kenyan Use (at least one) | How This Connects to the Anchoring Phenomenon. Students complete the row individually. Suggested correct entries: Family Name — Alkaline Earth Metals (Group 2); Key Properties — reactive but less so than Group 1, lose 2 electrons to form 2+ ions, silvery solids, react with acids to produce hydrogen gas, reactivity increases down the group; Electron Arrangement Pattern — 2 valence electrons in the outermost shell (e.g., Mg: 2,8,2 and Ca: 2,8,8,2); Kenyan Use — calcium in CAN fertiliser for Rift Valley maize farming, magnesium compounds in some antacid medicines sold in Kenyan pharmacies, calcium in cement used in SGR and Thika Road construction; How This Connects to the Anchoring Phenomenon — unlike sodium in the street lamp which reacts violently with water (1 outer electron), calcium in the fertiliser bag is safe to handle and	Teacher says: 'Open your cumulative chart to the next blank row. You have 4 minutes to fill in Group 2. Use your observation table, your balanced equations, and the electron diagrams we drew today. The Kenyan Use column must have at least one example you can explain — not just name.' [WAIT TIME: 4 minutes individual work — teacher circulates and checks charts.] Teacher looks specifically at the 'How This Connects' column — this is the hardest column and reveals whether students have made the link back to the phenomenon or are treating the chart as an isolated exercise. For any student who has left this column blank teacher prompts quietly: 'Think about the CAN fertiliser bag. Why is it safe to leave in the agrovet shop even when it rains? That is your connection.' In the final 60 seconds teacher says: 'Hold up your chart — show me the new row is filled. Next lesson you will need this chart, your element-sorting cards, and your observation table. Do not lose them.'	The cumulative reference chart is the core model-building artifact of the entire unit. Adding one row per lesson makes the growing explanatory power of chemical family thinking visible and tangible. The 'How This Connects to the Anchoring Phenomenon' column is a non-negotiable metacognitive prompt that prevents the lesson from becoming isolated content and keeps the storyline coherent across all eight lessons.	Teacher scans charts during the 4-minute window — priority check is the Electron Arrangement Pattern column (diagnostic for whether students can correctly draw and interpret electron shells) and the Kenyan Use column (diagnostic for whether contextual transfer has occurred). Students who have copied a Kenyan use without being able to explain it are flagged for a follow-up question at the start of Lesson 4: 'Last lesson someone wrote calcium in CAN fertiliser as a Kenyan use — can you tell me why calcium is in that fertiliser and not sodium?'

	dissolves slowly in soil moisture (2 outer electrons, harder to lose). Teacher ends the lesson by saying: 'Your model is growing. Every lesson we are adding a new chemical family. By Lesson 7 you will be able to explain everything in that anchoring phenomenon — the orange light, the argon welding shield, the fizzing tablet, and the helium balloon — using the model you have built yourself.'			

D. TEACHER REFLECTION

After the lesson teacher should consider: (1) Did the magnesium-HCl experiment produce clear results for all groups — if some groups saw no reaction, was this due to oxide layer on the ribbon or acid concentration? Adjust preparation for the next experiment lesson accordingly. (2) Did students correctly identify hydrogen gas via the splint test, or did some groups skip the gas test? If skipped, build the gas test explicitly into the Lesson 4 procedure card as a non-optional step. (3) Were students able to write balanced equations for $Mg + 2HCl \rightarrow MgCl_2 + H_2$ independently, or did most require significant scaffolding? If scaffolding was heavy, consider providing a worked example at the start of Lesson 4 as a revision anchor before introducing new equations. (4) Did the CAN fertiliser context land meaningfully — did students from farming backgrounds (especially those with family in the Rift Valley) engage more actively? If so, use their contextual knowledge as a teaching resource in the whole-class discussion in Lesson 7. (5) What did the yellow sticky notes on the DQB reveal about student thinking — are questions converging on halogens (expected) or diverging toward unrelated areas? Adjust the Lesson 4 opening to directly address the most common yellow-note question.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Magnesium ribbon reacted with dilute hydrochloric acid producing vigorous bubbling — hydrogen gas confirmed by squeaky pop with a lit splint — and the ribbon dissolved to form a colourless solution of magnesium chloride. Magnesium also reacted with steam (teacher demonstration) producing a white powder (magnesium oxide) and hydrogen gas, but the reaction with steam was slower than with acid. Magnesium did not react noticeably with cold water during the observation period.
What did I learn?	Alkaline earth metals (Group 2) have two valence electrons and form 2+ ions when they react. Having two electrons to remove requires more energy than removing one, so Group 2 elements are less reactive than Group 1. Reactivity increases down Group 2. Calcium and magnesium are used in CAN fertiliser for Kenyan farming because their controlled reactivity is agriculturally useful without being hazardous.
How does this explain the phenomenon?	The difference in reactivity between sodium (Group 1, one outer electron) and magnesium (Group 2, two outer electrons) is explained by the energy needed to remove outer electrons: removing two electrons is harder than removing one. This electron-arrangement pattern also explains why Group 2 elements can be safely used in fertiliser bags stored in agrovet shops and spread on Rift Valley maize fields — their moderate reactivity makes them useful, not dangerous. This advances the driving question: the number of valence electrons determines how reactive an element is, which determines what it can safely and usefully do in daily Kenyan life.

LESSON 4 (80 minutes (2 × 40-minute lessons)): Alkali Metals in Action: Why Does Reactivity Increase Down Group 1?	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	Students gather direct observational evidence of Group 1 (alkali metal) reactions with water, construct and balance chemical equations for Na + H_2O and K + H_2O, and explain the trend in reactivity down Group 1 using electron arrangement — advancing the class's ability to answer why the sodium street lamp on Thika Road behaves so differently from the argon welding shield or the helium party balloon.
Knowledge	Students will know that: (i) alkali metals (lithium, sodium, potassium) each have one outer-shell electron; (ii) reactivity increases down Group 1 as the outer electron is held less tightly by the nucleus; (iii) the reaction of alkali metals with water produces a metal hydroxide and hydrogen gas; (iv) balanced symbol equations for Na + H_2O and K + H_2O can be written and checked.
Skills	Students will be able to: (i) record structured observations using a Group 1 Observation Grid; (ii) write and balance chemical equations for alkali metal–water reactions; (iii) use electron configuration diagrams to explain the trend in reactivity; (iv) construct a cause–effect argument linking atomic structure to macroscopic behaviour; (v) update the class Driving Question Board with new evidence claims.
Attitudes	Students will appreciate that: the seemingly dramatic behaviour of sodium (as used in Thika Road street lamps) is a direct consequence of its atomic structure; patterns in the periodic table have predictive power; and safe, careful observation is the foundation of scientific explanation. Students demonstrate the value of Love — taking care of self and others — by observing all safety protocols during the demonstration.
Key Inquiry Question	How are elements suited to their use in day-to-day life?
Purpose in Storyline	Lessons 1–3 established that elements can be sorted into chemical families and that the periodic table organises them by electron arrangement. Lesson 4 is the first deep-dive into a specific family: Group 1. By watching sodium react violently with water, students gain visceral evidence that one outer electron = high reactivity, and they begin to build the atomic-structure explanation that will later distinguish Group 1 from Group 2 (Lesson 5) and from the completely unreactive noble gases (Lesson 6). The DQB is updated to bridge from alkali metals to alkaline earth metals.
Safety Notes	TEACHER DEMONSTRATION ONLY — students must remain seated at least 2 metres from the demonstration bench. Safety screen/blast shield must be in place. Teacher wears chemical-splash goggles, face shield, and nitrile gloves. Use only rice-grain-sized pieces of sodium and lithium stored under mineral oil; potassium demonstration is OPTIONAL and requires additional precautions (school must have a fire extinguisher rated for alkali metal fires). Water trough must be large and made of heat-resistant glass or a metal basin. In case of skin contact with alkali metal or sodium hydroxide solution, flush immediately with large amounts of water for at least 15 minutes and refer to first aid kit. Do NOT allow students to handle alkali metals. Dispose of alkali metal residues by careful addition of ethanol, not by discarding in water sink. Phenolphthalein indicator is used in small quantities — avoid skin and eye contact.

B. LESSON OVERVIEW

In this 80-minute lesson (split across two 40-minute periods), students first predict the behaviour of Group 1 metals with water, drawing on their knowledge of electron arrangement from Lesson 3. The teacher then performs a live demonstration (or shows a verified video clip) of lithium, sodium, and potassium reacting with water, and students record observations on a structured grid. Groups write and balance the equations for Na + H_2O and K + H_2O, then use electron configuration dot-diagrams to construct a causal argument explaining why reactivity increases down Group 1. The lesson closes with a DQB update that bridges to alkaline earth metals (Group 2) and sets up Lesson 5. Throughout, the anchoring phenomenon — sodium in Thika Road street lamps vs. argon in a welding shield — is kept in focus: the same property (one easily-lost outer electron) that makes sodium reactive in water also makes it emit characteristic orange-yellow light when electrically excited in a lamp tube.

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**
Phase 1 — PREDICT (15 minutes) 📹 **VIDEO:** [Naming ions and ionic compounds] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **PDF:** [Using group work in your classroom] Source: TESSA - Teacher Resources 🔍 Search ARES for similar readings	Working individually for 4 minutes, students write responses to two prompt questions in their ARES notebooks: (1) 'Sodium is used inside the street lamps along Thika Road. It has 1 outer-shell electron. What do you predict will happen if a small piece of sodium is dropped into a trough of water? Justify your prediction using what you know about electron arrangement.' (2) 'Potassium is directly below sodium in Group 1 and also has 1 outer-shell electron, but that electron is in a higher energy level (further from the nucleus). Do you predict potassium will react MORE or LESS vigorously than sodium? Why?' Students then share predictions with a shoulder partner for 2 minutes. The teacher collects 5–6 predictions via cold-call (targeting a mix of confident and hesitant students) and records them in two columns on the board — 'Predicts MORE vigorous' / 'Predicts LESS vigorous / same' — so the class can revisit them after observation. Students annotate their notebooks with a 'P' (Predict) timestamp.	SETUP: Before class, write the two predict prompts on the board or display them on the ARES screen. Have the Group 1 Observation Grid handout ready to distribute at the start of Phase 2. DURING PREDICT: Say — 'Before I show you anything, I want your brain's first guess. There are no wrong predictions — what matters is your reasoning.' Give 4 minutes of silent individual writing. Use a visible timer. After 4 minutes, say — 'Turn to your shoulder partner. You have 2 minutes. Tell each other your prediction AND your reason.' After 2 minutes, cold-call EXACTLY 6 students (pre-selected to represent range of confidence): 'Amina — what did you predict, and why?' WAIT 12 seconds for each response before moving on. After each response, do NOT evaluate — instead say 'Thank you — let's hold that idea and see what the evidence shows.' Record predictions on the board in the two columns. Close by saying — 'In about 10 minutes, you will be the judges of which predictions were closest to the evidence. Keep your notebooks open.'	PREDICT-OBSERVE-EXPLAIN (POE) Launch — By committing predictions to paper before seeing evidence, students activate prior knowledge (electron arrangement from Lesson 3) and create a personal cognitive stake in the outcome. Recording predictions publicly on the board creates accountability and makes conceptual change visible when observations contradict predictions.	Observable indicator: Every student has written at least one sentence of prediction AND one sentence of justification referencing electron arrangement in their notebook before cold-call begins. Teacher scans notebooks during partner talk (walk past 4–5 desks). RED FLAG: If fewer than half of students mention 'electron' or 'outer shell' in their justification, pause and ask — 'What did we learn in Lesson 3 about why elements in the same group have similar properties?' before proceeding to observation.
Phase 2 — OBSERVE (25 minutes) 📹 **VIDEO:** [Naming ions and ionic compounds] Source: Khan Academy (English - US curriculum) 🔍 Search ARES	Teacher distributes the Group 1 Observation Grid (pre-printed or drawn in notebooks) with columns: Metal | Physical appearance | Behaviour in water (speed, fizzing, movement, flame?) | Colour of phenolphthalein solution after reaction | Balanced equation. Students watch the teacher	SAFETY BRIEFING (2 min): Before any demonstration, say — 'Everyone seated, two metres back. Goggles on. No one approaches the bench until I say so. This is real chemistry — the same element that makes the Thika Road lamps glow orange is about to meet water.' Check all	STRUCTURED OBSERVATION GRID + CLAIM-EVIDENCE REASONING — The grid scaffolds students to move from raw observation ('it fizzed and moved around') to a patterned claim ('reactivity increases down the group'). This is NGSS Science and Engineering Practice 3 (Planning	Observable indicators: (1) Each student's Observation Grid has at least 3 of 4 columns completed for each metal. (2) Each group's mini-whiteboard shows a balanced symbol equation for Na + H_2O ($2Na + 2H_2O \rightarrow 2NaOH + H_2$) — teacher checks for correct coefficients and the H_2 product. (3)

for similar videos 📖 **PDF:** Using group work in your classroom Source: TESSA - Teacher Resources 🔍 Search ARES for similar readings	demonstrate, in sequence: (A) lithium in water, (B) sodium in water, (C) potassium in water (or a verified KNEC-approved video clip of potassium if live demo is not permitted by school safety policy). For each metal, students first watch silently for 30 seconds, then complete the row of the Observation Grid. After all three demonstrations, groups of 4 discuss: 'What pattern do you see in the observations as you go from Li → Na → K?' Groups record a one-sentence group claim. Two groups share with the class. Teacher writes the agreed observation summary on the board: 'Reactivity INCREASES going down Group 1: Li < Na < K.' Students also observe that the water turns pink with phenolphthalein, confirming a hydroxide (alkaline product) is formed. In the second half of Phase 2, student groups attempt to write and balance the equations for Na + H_2O and K + H_2O on mini-whiteboards. Teacher circulates and uses targeted questioning (see Teacher Moves).	students are seated and goggles are on. DEMONSTRATION SEQUENCE: For each metal, narrate what you are doing: 'I am cutting a rice-grain-sized piece of lithium from under its mineral oil storage. Watch what happens the moment it touches water.' After Li: 'Write what you saw — be specific. Fizzing? Movement? Flame?' Give 60 seconds of writing time. After Na: same narration and writing pause. After K (live or video): same. PATTERN DISCUSSION: After all three, ask — 'What is different about the potassium reaction compared to the lithium reaction?' WAIT 12 seconds. Cold-call 3 students. Push for specificity: 'You said it was more violent — can you describe exactly what was more violent?' EQUATION BUILDING: Say — 'Now, in your groups, write the word equation for sodium reacting with water, then try the symbol equation. You have 4 minutes.' Circulate. When a group has a correct unbalanced equation, ask — 'How do you know if this equation is balanced? Count the atoms on each side.' For groups struggling, ask — 'What are the PRODUCTS? Hint — the pink colour tells you one product. What do you think the fizzing gas is?' AFTER MINI-WHITEBOARD SHARE: Choose one board with the correct balanced equation ($2Na + 2H_2O \rightarrow 2NaOH + H_2$) and display it. Ask — 'Why is there a coefficient of 2 in front of sodium and water?' WAIT 10 seconds. Cold-call 2 students.	and Carrying Out Investigations) and Practice 4 (Analysing and Interpreting Data). The mini-whiteboard equation task applies Practice 5 (Using Mathematics and Computational Thinking) to represent the chemical change symbolically.	Groups can state the reactivity trend verbally. RED FLAG: If groups write $NaOH_2$ or omit H_2 as a product, stop the class and return to the word equation: 'What did the pink colour prove? What was the gas?' Recheck before moving to Phase 3.
Phase 3 — EXPLAIN (20 minutes)	Students are given a 'Group 1 Electron Structure Card' showing the dot-cross diagrams for Li (2,1),	DISTRIBUTE Electron Structure Cards. Say — 'You now have the atomic portraits of three Group 1	GUIDED QUESTION SCAFFOLDING + CONSENSUS EXPLANATION BUILDING — The	Observable indicators: (1) Each group's written explanation contains the words 'outer electron',

📹 **VIDEO:** [Naming ions and ionic compounds] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **PDF:** [Using group work in your classroom] Source: TESSA - Teacher Resources 🔍 Search ARES for similar readings	Na (2,8,1), and K (2,8,8,1). Working in groups of 4, students answer three guided sensemaking questions on the card: (Q1) 'In each of these atoms, how many outer-shell electrons are there?' (Q2) 'As you go from Li → Na → K, what happens to the distance of the outer electron from the nucleus? What does this mean for how tightly the nucleus holds that electron?' (Q3) 'Use your answer to Q2 to explain why potassium reacts more vigorously with water than lithium does.' Groups construct a written Group Explanation (3–5 sentences). One spokesperson per group reads their explanation to the class. Teacher synthesises the best elements of group explanations into a CLASS CONSENSUS EXPLANATION written on the board: 'All Group 1 metals have 1 outer-shell electron. Going down the group, each successive element has an extra electron shell, so the outer electron is further from the nucleus and is held less tightly. This means the outer electron is more easily lost, making the element more reactive. That is why K reacts more vigorously than Na, which reacts more vigorously than Li.' Students copy this into their notebooks and annotate it with arrows showing increasing reactivity. Teacher then returns to the anchoring phenomenon: 'So — the sodium in the Thika Road street lamp has this one outer electron that is quite easily excited. When electricity passes through sodium vapour in the lamp, that outer electron absorbs energy and then releases it as orange-yellow light. The very property that makes sodium	metals. Your job as a group is to write a scientific explanation — not just a description — for the pattern you observed. A scientific explanation answers WHY.' Give groups 8 minutes to work through Q1–Q3 and write the group explanation. Circulate — do NOT give answers. Use prompts: 'You wrote that the outer electron is further away — further away means the attraction is stronger or weaker? Think about gravity — does the Earth pull harder on something close or something far?' After 8 minutes, cold-call 3 groups to share. For each, ask the class — 'Do you agree? Does their explanation actually answer WHY reactivity increases?' WAIT 10 seconds. After three groups have shared, build the CLASS CONSENSUS EXPLANATION on the board live, saying — 'I am going to write what I am hearing from all of you. Tell me if I am missing anything.' Write the 4-sentence consensus. PHENOMENON RETURN (3 min): Return to the anchoring phenomenon image/slide (Thika Road street lamps + welding torch + balloon). Say — 'We just explained why sodium is so reactive. But inside that street lamp, sodium is sealed in a glass tube with no water or air. It cannot react chemically. Yet it glows orange. How can the SAME property — one loosely held outer electron — explain BOTH the violent water reaction AND the beautiful orange glow? Discuss with your partner for 90 seconds.' Cold-call 2 pairs. Affirm the key idea: 'Yes — the outer electron is both easily lost in reactions AND easily excited by energy. Same	three graduated questions on the Electron Structure Card apply NGSS Practice 6 (Constructing Explanations) by moving students from data (electron configurations) to a mechanism (nuclear attraction decreases with distance) to a prediction (K more reactive than Na). The teacher's live co-construction of the consensus explanation models scientific argumentation and ensures all students leave with a correct, complete explanation in their notebooks.	'distance from nucleus', and 'more easily lost' (or equivalent phrasing). (2) During cold-call, students can articulate the causal chain (more shells → outer electron further → weaker attraction → more easily lost → more reactive) without reading directly from the card. (3) At least 2 students can connect the 'easily excited outer electron' idea back to the Thika Road phenomenon unprompted. RED FLAG: If groups are still writing purely descriptive answers ('potassium is more reactive because it is lower in the group'), probe with — 'That describes the pattern — now explain the cause. What is happening at the atomic level?'

	reactive with water — that loosely held outer electron — is ALSO what makes it glow orange. How does this help us answer our driving question?'	electron, two phenomena.'		
Phase 4 — DRIVING QUESTION BOARD (DQB) Update (10 minutes) 📹 **VIDEO:** [Naming ions and ionic compounds] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **PDF:** [Using group work in your classroom] Source: TESSA - Teacher Resources 🔍 Search ARES for similar readings	Each student receives two sticky notes (or writes in the ARES DQB digital interface): one GREEN ('We now know…') and one ORANGE ('We still wonder…'). Students write individually for 3 minutes, then post notes on the class DQB under the section labelled 'Group 1 — Alkali Metals'. The teacher reads 4–5 notes aloud, grouping the 'We now know' notes under the claim: 'Alkali metals lose 1 outer electron easily — reactivity increases down Group 1.' The teacher then highlights two or three 'We still wonder' notes that preview upcoming lessons — particularly questions about alkaline earth metals (Group 2) — and writes the bridge question on the DQB in a new column: 'We now know alkali metals lose 1 electron easily — but why do alkaline earth metals (Group 2, with 2 outer electrons) behave differently? Will they be more or less reactive?' Students record this bridge question in their notebooks under 'Next Investigation.'	Distribute sticky notes (or open ARES DQB interface). Say — 'Two notes. Green: one thing you are NOW confident you know and can explain — not just describe. Orange: one thing you are still genuinely wondering about. You have 3 minutes — go.' After 3 minutes, collect and post notes. Read GREEN notes aloud and cluster them: 'I am seeing a theme — many of you are writing about the outer electron and distance from nucleus. That is our new knowledge anchor.' Read 3–4 ORANGE notes aloud: 'Several of you are asking about Group 2 — magnesium, calcium. Great scientific instinct. If Group 1 has 1 outer electron and is very reactive, what do you predict Group 2 will be like with 2 outer electrons?' WAIT 12 seconds. Do NOT answer — say — 'Hold that question. It goes on the DQB right here.' Write the bridge question on the DQB visibly. Say — 'Every lesson, we are building a bigger picture. Next lesson, we test that prediction with magnesium.' Point back to the anchoring phenomenon: 'And remember — our goal is still to explain why the Thika Road lamp, the welding torch, the pool chlorine tablet, and the helium balloon ALL behave differently, even though they are all made of elements. We have just explained one piece of that puzzle.'	DRIVING QUESTION BOARD — Evidence Anchoring + Storyline Progression — The DQB functions as a cumulative public record of the class's growing scientific understanding (NGSS Practice 7: Engaging in Argument from Evidence). By separating 'We now know' from 'We still wonder', students practise distinguishing claims supported by evidence from open questions — a core epistemic habit of science. The teacher's explicit bridging move connects today's evidence to tomorrow's investigation, maintaining the coherence of the 8-lesson storyline.	Observable indicators: (1) Every student has posted at least one GREEN and one ORANGE sticky note. (2) GREEN notes reference specific evidence from today's lesson (e.g., 'Na reacts faster than Li', 'K produced a flame', 'the outer electron is loosely held') — not generic statements like 'I learned about Group 1.' (3) At least one student's ORANGE note anticipates Group 2 behaviour, suggesting transfer of the 'outer electrons' framework. RED FLAG: If GREEN notes are all vague ('I learned chemistry today'), the teacher should say — 'I need you to be more specific. Finish this sentence on a new note: The reason potassium is more reactive than sodium is that ______.'
Phase 5 —	Students retrieve their cumulative	Say — 'Get out your Chemical	CUMULATIVE MODEL REVISION	Observable indicators: (1) Each

MODEL BUILDING (10 minutes) 📹 **VIDEO:** Naming ions and ionic compounds Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **PDF:** Using group work in your classroom Source: TESSA - Teacher Resources 🔍 Search ARES for similar readings	'Chemical Families Model' — a large annotated diagram they have been building across all 8 lessons. In Lessons 1–3, they added: the periodic table layout, electron configuration rules, and the concept of chemical families. In Lesson 4, students add a NEW PANEL to their model labelled 'Group 1 — Alkali Metals.' In this panel they draw: (a) the electron configuration diagrams for Li, Na, K (dot-cross style); (b) an arrow pointing DOWN the group labelled 'increasing reactivity'; (c) the balanced equation $2Na + 2H_2O \rightarrow 2NaOH + H_2$ with annotations (products: lye/caustic soda + flammable hydrogen gas); (d) a small annotated sketch of a sodium street lamp with the label 'loosely held outer electron → absorbs and emits energy as orange-yellow light.' Students add a dashed BRIDGE ARROW from Group 1 to a blank Group 2 panel (to be completed in Lesson 5), labelled with the bridge question from the DQB. With 2 minutes remaining, the teacher leads a whole-class 60-second 'model tour': two volunteers hold up their models and narrate one addition they made today.	Families Model from your portfolio. You are going to add today's evidence to it. You have 7 minutes. Your model must show: the electron configs for Li, Na, K; the reactivity arrow; one balanced equation; and the sodium lamp connection to our phenomenon. You can also add colour — sodium = orange!' Circulate and check models. Prompt students who are only copying equations without annotations: 'Why does the reactivity arrow point DOWN? Write that explanation next to the arrow — do not just draw it.' At 7 minutes, say — 'Models down. Two volunteers — hold yours up. Give me a 30-second tour of what you added today.' After 2 volunteers, close with — 'Every lesson, your model gets smarter. By Lesson 8, your model will explain why the sodium lamp, the argon welding shield, the pool chlorine tablet, and the helium balloon are ALL built the way they are. Today was Group 1. Next: Group 2.'	— NGSS Practice 2 (Developing and Using Models) — The cumulative model is not a drawing exercise but an epistemic tool: it externalises student thinking and makes growth in understanding visible across the 8-lesson sequence. By requiring both symbolic representations (equations, electron configs) and narrative annotations (explanations in the student's own words), the model bridges formal chemistry notation and conceptual understanding. The BRIDGE ARROW creates a visual anticipation structure that primes students for Lesson 5.	student's model has all four required additions (electron config diagrams, reactivity arrow with direction, balanced equation with annotation, sodium lamp phenomenon link). (2) The reactivity arrow is labelled with a causal explanation, not just 'increases going down.' (3) The bridge arrow to Group 2 is present with the question written out. EXIT CHECK: As students pack up, teacher stands at the door and asks each student one of three rotating questions: (Q-A) 'What product besides NaOH is formed when sodium reacts with water?' (Q-B) 'Why is potassium more reactive than sodium?' (Q-C) 'What property of sodium explains both its reaction with water AND its orange glow in a lamp?' Students answer in 1–2 sentences before leaving. Teacher tallies correct responses to gauge class readiness for Lesson 5.

D. TEACHER REFLECTION

After delivering this lesson, reflect on the following questions and record responses in your ARES Teacher Journal:

1. PREDICTION QUALITY: Did students' initial predictions reference electron arrangement (evidence of transfer from Lesson 3), or were predictions based purely on prior experience (e.g., 'I've seen sodium explode on YouTube')? If predictions lacked atomic-structure reasoning, consider adding a 5-minute electron configuration warm-up at the start of Lesson 5 before the Group 2 evidence activity.

2. OBSERVATION RIGOUR: Were students recording specific, measurable observations (e.g., 'potassium moved across the surface and produced a purple-violet flame') or vague descriptions ('potassium was more violent')? If descriptions were vague, introduce the 'Observable vs. Interpretable' distinction at the start of Phase 2 in

Lesson 5: 'An observation is something your senses detect directly. An interpretation is what you conclude from it.'

3. EQUATION BALANCING: How many groups correctly balanced $2Na + 2H_2O \rightarrow 2NaOH + H_2$ on the first attempt? If fewer than 60% of groups balanced correctly, build a 5-minute equation-balancing drill into the start of Lesson 5, using the $Mg + H_2O$ equation as practice.

4. PHENOMENON CONNECTION: Did students spontaneously connect the 'loosely held outer electron' to the orange lamp glow during Phase 3, or did this connection require significant teacher scaffolding? If scaffolding was heavy, consider displaying the anchoring phenomenon image more prominently throughout the lesson (not just in Phase 3) to keep the cognitive thread alive.

5. DQB BRIDGE QUESTIONS: Read the ORANGE sticky notes carefully before Lesson 5. Do students' 'We still wonder' questions cluster around Group 2 reactivity (target), or are they still wrestling with Group 1 concepts (gap)? Adjust the opening of Lesson 5 to address unresolved Group 1 confusions before introducing new evidence.

6. SAFETY: Record any near-misses or safety concerns during the demonstration. If the school's safety policy prohibits live alkali metal demonstrations, identify a verified Kenyan-accessible video source (e.g., Kenya Education Cloud, KNEC-approved resource) for use in Lesson 4 revision or for schools without laboratory facilities. Ensure alkali metal storage and disposal procedures comply with the school's chemical safety register.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Students watched a teacher demonstration (or verified video) of lithium, sodium, and potassium reacting with water and recorded observations on a structured Group 1 Observation Grid — noting increasing speed of reaction, fizzing (hydrogen gas), movement on the water surface, and the production of an alkaline solution confirmed by phenolphthalein turning pink. The observation that potassium produced a lilac/violet flame while sodium did not was noted as a key qualitative difference.
What did I learn?	Students learned that all Group 1 alkali metals have exactly 1 outer-shell electron; that this outer electron is lost more easily as atomic radius increases going down the group (because the outer electron is further from the nucleus and shielded by more inner electron shells, reducing the effective nuclear attraction); that this explains why reactivity increases $Li < Na < K$; and that the reaction of any alkali metal with water follows the pattern: metal + water → metal hydroxide + hydrogen gas, with balanced equations $2Na + 2H_2O \rightarrow 2NaOH + H_2$ and $2K + 2H_2O \rightarrow 2KOH + H_2$. Students also connected sodium's loosely held outer electron to its behaviour in street lamp tubes — the same electron property drives both chemical reactivity and the characteristic orange-yellow light emission.
How does this explain the phenomenon?	Students constructed a class consensus causal explanation: 'All Group 1 metals have 1 outer-shell electron. Going down the group, each successive element has one more electron shell, placing the outer electron further from the nucleus. The electrostatic attraction between the nucleus and the outer electron therefore decreases, making the electron more easily lost. A more easily lost outer electron means the metal reacts more vigorously with water, producing metal hydroxide and hydrogen gas faster. This is why potassium reacts most vigorously of the three, sodium intermediate, and lithium least vigorously — and it is why sodium inside a Thika Road street lamp tube, though not reacting chemically, readily absorbs electrical energy through that same loosely held outer electron and re-emits it as distinctive orange-yellow light.'

LESSON 5 (40 minutes): Noble Gases: Why Do Some Elements Do Absolutely Nothing?	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	Students will explain why noble gases are chemically inert by connecting their full outer electron shells to their stability, and will link each noble gas to a specific Kenyan use that depends on that inertness.
Knowledge	Students know that Group 18 elements (noble gases) have completely filled outer electron shells (helium: 2; neon, argon, krypton, xenon: 8); that a full outer shell means no tendency to gain, lose, or share electrons; that this electron arrangement explains chemical inertness; and that noble gases are colourless, odourless, monatomic gases at room temperature with very low boiling points.
Skills	Students can research and record properties and uses of noble gases using digital devices or print media; construct a structured comparison chart linking each noble gas to its outer-shell electron count, key property, and a Kenyan use; and communicate findings in a group mini-presentation.
Attitudes	Students appreciate that stability and non-reactivity are as scientifically significant as dramatic reactivity, and develop curiosity about how an element that does nothing chemically can still be extremely useful in everyday Kenyan life.
Key Inquiry Question	How are elements suited to their use in day-to-day life? — Noble gases are suited to uses that require a chemically inert, non-flammable, non-toxic environment precisely because their full outer shells mean they will not react with any surrounding material.
Purpose in Storyline	Lesson 5 resolves the biggest puzzle students posted on the Driving Question Board after Lesson 4: how can an element that does absolutely nothing still be useful? By anchoring the explanation in the argon welding shield from Kamukunji and the helium weather balloon from the Kenya Meteorological Department, students see that inertness IS the functional property — completing the reactive-to-inert spectrum before Lesson 6 introduces the variable, colourful world of transition metals.
Safety Notes	

B. LESSON OVERVIEW
Students begin by revisiting the anchoring phenomenon — the argon-shielded jua kali welding flame and the helium party balloon — and are challenged to explain why these gases behave so differently from the alkali metals and halogens studied in Lessons 2–4. Through a structured predict-observe-explain cycle, students first record predictions about what makes noble gases special, then conduct a guided research activity (digital or print) to discover outer-shell electron counts for He, Ne, Ar, Kr, and Xe. Groups construct a mini-chart linking each noble gas to its outer-shell configuration, its key inert property, and a verified Kenyan use. The lesson closes with students updating the class Driving Question Board, articulating that full outer shells explain both chemical inertness and specific industrial utility, and generating a bridging question toward transition metals in Lesson 6.

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**
Predict Phase 📹 **VIDEO:** Kinetic molecular	Students are shown two projected images side by side: (1) a jua kali welder on River Road, Nairobi,	Display both images simultaneously and say: 'You have seen sodium react explosively with	Prediction journaling followed by turn-and-talk activates prior knowledge of electron	Teacher circulates during the 90-second silent write and reads at least 6 notebooks. Checkpoint: Do

theory of gases Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Noble Gas Configuration (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	surrounded by a faint shimmer of argon gas shielding his torch, and (2) a Kenya Meteorological Department technician in Dagoretti releasing a large helium weather balloon. The teacher asks students to silently write in their notebooks for 90 seconds: 'What do argon and helium have in common? Why would a welder WANT a gas that does nothing?' Students then turn-and-talk with a partner for 60 seconds before three pairs share with the class. The teacher records key prediction words on the board — students typically suggest 'does not burn', 'safe', 'does not react', 'stable' — without correcting or affirming. Students are then shown the DQB sticky note from Lesson 4 that reads 'Noble gases are stable — but WHY?' and told this lesson will answer that question using electron arrangements they already know from Sub-Strand 2.1.	water. You have seen chlorine bleach a flower. Now look at these two uses of elements. The welder NEEDS the gas to protect his work — but the gas must not react with anything. The meteorologist NEEDS the balloon to rise — but it must not burn or explode near instruments. Take 90 seconds — write your best prediction: what property does a gas need to be useful in BOTH these situations?' [WAIT 90 SECONDS — circulate silently, read notebooks, note 2-3 interesting predictions to draw out]. After turn-and-talk, call on pairs: 'Achieng, what did your pair write?' After three responses, say: 'Notice — every pair used the word react or safe or stable. Today we are going to find out WHY noble gases have that property, and connect it back to the one idea that explains everything in this unit: electron arrangement.'	configuration from Sub-Strand 2.1 and creates a concrete need-to-know before new information is introduced. Anchoring predictions to the DQB question from Lesson 4 shows students their own questions are driving the lesson.	students connect inertness to electron arrangement, or only to physical properties like 'gas'? Students who only cite physical state are prompted: 'Think about what we said about Group 1 and Group 17 — what did their outer electrons do? What might happen if the outer shell were already full?'
Observe Phase 📹 **VIDEO:** Kinetic molecular theory of gases Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Noble Gas Configuration (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Groups of four receive a structured research card with the following task: 'Using your textbook pages on Group 18, the periodic table poster, and/or the approved websites on your device, find and record in the table below: (a) the name and symbol of each noble gas He, Ne, Ar, Kr, Xe; (b) the number of electrons in the outer shell of each; (c) one physical property (colour, state at room temperature); (d) one real-world use in Kenya or East Africa.' Students have 10 minutes for the research task. The teacher displays a partially completed model row on the board: 'Neon | Ne | 8 outer electrons | colourless gas | orange-red glow in road signage along Mombasa Road commercial buildings.' While	Distribute research cards and say: 'You have 10 minutes. Your job is to be a detective — find the pattern in the outer shells. I am not going to tell you the answer yet. Use the periodic table, your textbook Group 18 section, or the two approved links on the board.' [WAIT — circulate to each group, spending no more than 90 seconds per group]. After 7 minutes say: 'Two more minutes — make sure every member of your group can explain the outer-shell column, not just the person who wrote it.' After 10 minutes, pause groups and show the video clips: 'Watch these 60 seconds. Add anything new you observe to your research card.' After the clips, ask: 'Raise your hand if you noticed something in the outer-	Guided research with a structured recording table ensures all groups collect parallel data, making cross-group comparison possible. The partially completed model row scaffolds without removing the discovery element. The video clips provide observational anchoring so the research does not remain purely abstract.	Collect one research card per group at the end of the phase. Check: (1) Is the outer-shell electron count correct — 8 for Ne, Ar, Kr, Xe and 2 for He? (2) Has the group identified at least one Kenyan or East African use per gas? (3) Is the use linked to inertness or a physical property, not merely listed? Groups with incorrect outer-shell counts receive a targeted prompt card: 'Count the electrons in period 2 of your periodic table poster — how many are in the last shell of neon?'

	groups work, the teacher projects a 30-second silent video clip of a neon sign flickering to life on a Nairobi street and a second 30-second clip of a jua kali welder with visible argon shielding. Students annotate their research cards as they observe. Groups that finish early are asked: 'Find out what the outer shell of helium looks like — is it the same number as the others? Why might that still count as full?'	shell column that is the same for almost all noble gases.' [WAIT 10 SECONDS]. Call on a student: 'Wambua, what did you notice?' Use the response to bridge to the Explain phase.		
Explain Phase 📹 **VIDEO:** [Kinetic molecular theory of gases] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Noble Gas Configuration (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	The teacher draws a large electron shell diagram for argon (2, 8, 8) on the board beside a diagram for potassium (2, 8, 1) and chlorine (2, 8, 7) — both already familiar from previous lessons. Students are asked: 'In your notebook, write one sentence explaining why argon cannot react the way potassium does and cannot react the way chlorine does.' After 60 seconds, students share sentences with the class. The teacher formalises the explanation: a full outer shell means no energetic need to gain, lose, or share electrons, so no chemical bond can form under ordinary conditions. Students write the key principle: 'Noble gases are inert because their outer electron shells are completely full — 2 for helium, 8 for all others — giving them no tendency to gain, lose, or share electrons.' The teacher then facilitates a whole-class mini-presentation round: each group reads out their noble gas, its outer-shell count, its Kenyan use, and one sentence explaining why that use depends on inertness. The teacher annotates the board chart as groups present, building a collective reference. Students correct and complete their own	After drawing the three shell diagrams, say: 'Potassium had one outer electron and reacted explosively with water. Chlorine had seven outer electrons and grabbed one more to bleach a flower. Argon has eight outer electrons in its outer shell. Take 60 seconds — write one sentence: why can argon not do what potassium did, and why can it not do what chlorine did?' [WAIT 60 SECONDS]. Call on three students with different phrasings. Then say: 'All of you are describing the same idea with different words. The scientific term is chemical inertness, and it comes entirely from this: the outer shell is full. Nothing to give. Nothing to take. Nothing to share.' Write on the board: FULL OUTER SHELL = CHEMICAL INERTNESS. Then transition to group presentations: 'Each group will now present their noble gas in exactly 60 seconds. I want to hear: name, outer electrons, one Kenyan use, and one sentence linking the use to inertness. Karibu — who wants to go first?' After each presentation, ask the class: 'Does the rest of the class agree that this use depends on inertness? Thumbs up if yes,	Comparative electron shell diagrams (Ar vs K vs Cl) make the contrast between full and incomplete outer shells visually explicit, directly connecting new content to prior-lesson knowledge. The 60-second per group presentation format creates low-stakes peer teaching and ensures every student hears five different Kenyan use cases with explanations. The thumbs check provides real-time comprehension data.	Listen for the quality of the explanatory sentence linking use to inertness. Target misconception: students may say 'argon is used in welding because it is a gas' — redirect with: 'Many gases exist — why not use oxygen or nitrogen? What would happen if the welder used chlorine gas instead of argon?' This forces students to articulate inertness as the functional property, not just physical state. Record which groups can articulate the electron-structure explanation versus those who only describe the use.

	research cards based on peer input. Teacher explicitly connects helium (2 outer electrons, shell 1 is full at 2) to the question raised in the Observe phase, explaining that for period-1 elements the first shell is complete at 2 electrons, not 8.	thumbs sideways if you are not sure.' Address any thumbs-sideways responses immediately.		
Driving Question Board (DQB) Creation 📹 **VIDEO:** Kinetic molecular theory of gases Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Noble Gas Configuration (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Students return to the class DQB. The teacher points to the Lesson 4 sticky note: 'Noble gases are stable — but WHAT makes them stable, and how can something that does nothing still be useful?' Each student writes one sentence on a sticky note beginning with 'We now know...' and places it over the original question. Examples expected: 'We now know noble gases are stable because their outer shells are completely full, leaving nothing to gain, lose, or share.' 'We now know helium is used in Kenya Meteorological Department weather balloons because it cannot catch fire like hydrogen would.' Students then scan the remaining unanswered DQB questions and the teacher reveals the bridging question printed on a new card: 'Noble gases are stable, Group 1 metals are reactive, halogens grab electrons — so where do copper, iron, and zinc fit? They are metals but they behave completely differently from Group 1 — WHY?' This new card is posted on the DQB to preview Lesson 6.	Walk to the DQB and say: 'Four lessons ago you asked why some elements do nothing at all. You now have the answer. Take a sticky note — write one sentence starting with the words WE NOW KNOW. Put it directly over the question it answers.' [WAIT 2 MINUTES for writing and posting]. Read three posted notes aloud, paraphrasing each: 'Amina writes — we now know noble gases do nothing because their outer shells are full. Excellent — she has connected behaviour to structure.' Then post the new bridging card and say: 'This is your next question. Copper wire carries electricity across Kenya. Iron makes the SGR rails. Zinc coats the mabati on your roof. These are metals — but they are NOT like sodium or potassium. In Lesson 6 you will find out why. Add any sub-questions you have about transition metals to the DQB before you leave today.'	The DQB sticky-note replacement ritual makes visible the intellectual progress of the class — students physically cover old questions with new knowledge, creating a tangible record of evidence accumulation. The bridging question technique maintains narrative continuity in the storyline and creates anticipatory curiosity for Lesson 6.	Read all posted sticky notes after class. Check that students are writing explanatory statements (linking outer shells to inertness) not merely descriptive ones (noble gases do not react). Students who post only descriptive notes receive individual written feedback: 'Good observation — now add one sentence explaining WHY using the words outer shell and electrons.' Count how many students spontaneously add transition metal sub-questions to the DQB — this indicates the degree to which the bridging question has activated genuine curiosity.
Model Building Phase 📹 **VIDEO:** Kinetic molecular theory of gases Source: Khan	Each student individually completes a Model Card (half-A4 sheet provided) with four columns: Noble Gas Name | Outer Electron Count | Key Property That Makes It Useful | One Kenyan Use. Students fill in all five noble gases	Distribute Model Card sheets and say: 'This is your personal scientific model. Cover your research card. Fill in the table from memory first — this tells you what you have actually learned, not what you copied. After two	The memory-first then correct-in-colour approach is a metacognitive strategy that helps students distinguish secure knowledge from recently acquired information — building self-regulation skills aligned with CBE competencies.	Collect Model Cards at the door as an exit ticket. Evaluate: (1) Are outer electron counts correct for all five noble gases, including He = 2? (2) Does the model statement include a causal explanation referencing outer shell? (3) Does

Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Noble Gas Configuration (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	(He, Ne, Ar, Kr, Xe) from memory first, then check against their research cards and correct in a different colour pen. Below the table, students write a two-sentence model statement: 'Noble gases are chemically inert because [BLANK]. This makes argon suitable for jua kali welding in Kamukunji because [BLANK].' Students who complete both tasks early add a third sentence: 'If a welder mistakenly used chlorine gas instead of argon as a shield, the result would be [BLANK] because [BLANK].' The completed Model Cards are retained by students in their science portfolios as a reference for Lesson 7 (whole-unit poster) and Lesson 8 (cumulative reference chart).	minutes, uncover your research card and correct in a different colour. Two colours show you what you knew and what you still need to consolidate.' [WAIT 3 MINUTES for independent work]. After 3 minutes: 'Now write the two model sentences at the bottom. Use the words outer shell and inert — both must appear in your sentences.' [WAIT 2 MINUTES]. Circulate and provide targeted verbal feedback. For early finishers, prompt the extension question verbally: 'Wanjiku, you are done — here is a harder question: what would happen if the jua kali welder used chlorine instead of argon? Write your prediction in the extension box.' Close the phase by saying: 'Keep this card safe in your portfolio. In Lesson 7 you will use it to build the big class poster connecting all chemical families. In Lesson 8 it becomes part of your final model.'	The two-sentence model template enforces causal reasoning (because...) rather than mere description. The extension question uses a counterfactual to deepen understanding of why inertness — not just being a gas — is the functional property.	the Kenyan use correctly match the gas (not arbitrary)? Sort cards into three piles — secure, developing, needs support — and use this data to form targeted review groups at the start of Lesson 6 before introducing transition metals.

D. TEACHER REFLECTION

After the lesson, consider: Did students move from describing noble gas inertness (they do not react) to explaining it (full outer shell leaves nothing to gain, lose, or share)? Were Kenyan uses genuinely linked to the inert property, or were students just listing facts? Did the comparative shell diagrams (Ar vs K vs Cl) make the electron-structure argument clear, or do students need a further visual consolidation at the start of Lesson 6? Check the exit-ticket Model Cards — if more than one-third of students have He listed with 8 outer electrons, begin Lesson 6 with a 3-minute targeted correction on period-1 shell capacity. Also reflect on pacing: the research activity and group presentations are the most time-sensitive phases — if either ran over, consider whether the research card structure needs simplification or whether devices caused access delays that can be pre-empted next lesson.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Students explained the full outer shell principle using comparative electron shell diagrams for Ar (2,8,8), K (2,8,1), and Cl (2,8,7), articulating that Group 1 and Group 17 elements react because their outer shells are incomplete — one short or one excess — while noble gases have no such incompleteness. Students connected this structural explanation to the anchoring phenomenon: argon in the jua kali welding shield does not react with the molten metal or atmospheric oxygen because its outer shell is full, making it an ideal inert shield gas. Students updated the Driving Question Board, replacing the Lesson 4 unanswered question with evidence-based explanations, and generated a new bridging question about transition metals to anchor Lesson 6.

What did I learn?	Students learned that a full outer electron shell produces chemical inertness — no energetic driving force exists for a noble gas to form bonds under ordinary conditions. They learned that this inertness is the exact property exploited in each industrial use: argon shields molten metal from oxidation precisely because it will not react with the metal or the oxygen present; helium lifts weather balloons safely precisely because it cannot combust the way hydrogen would. Students also learned that helium is anomalous in having only 2 outer electrons because the first electron shell holds a maximum of 2, making it nonetheless complete.
How does this explain the phenomenon?	

LESSON 6 (80 minutes (2 × 40-minute lessons)): Magnesium in Action: Group 2 Reactivity, Fertilisers, and the Two-Electron Difference

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	To investigate the reactivity of magnesium (a Group 2 alkaline earth metal) with dilute hydrochloric acid and with water, compare its behaviour to Group 1 metals, and connect Group 2 properties to real Kenyan agricultural and industrial uses.
Knowledge	Learners will be able to: (a) identify chemical families in the periodic table; (b) classify elements of the periodic table into their chemical families; (c) investigate the physical and chemical properties of selected elements of the periodic table; (d) outline the uses of elements of the periodic table; (e) appreciate the uses of the elements of the periodic table in day-to-day life.
Skills	Carrying out a supervised experiment on the reaction of magnesium with dilute HCl and with cold/hot water; recording observations systematically; writing and balancing chemical equations for Mg + HCl and Mg + H_2O(steam); comparing reactivity trends across Group 1 and Group 2; interpreting data to draw evidence-based conclusions.
Attitudes	Self-efficacy: as the learner carries out activities to investigate the reaction of magnesium with dilute acids and water. Love: as the learner takes care of self and others by observing safety when carrying out activities to investigate properties of elements of the periodic table and appropriately using first aid kits in case of injuries.
Key Inquiry Question	How are elements suited to their use in day-to-day life?
Purpose in Storyline	In Lessons 1–5 students identified chemical families, explored Group 1 alkali metal reactivity (sodium exploding in water), investigated halogen and noble gas properties, and began sorting elements by family. Lesson 6 deepens the Group 2 investigation: students directly test magnesium — a safer, slower-reacting Group 2 metal — and generate their own experimental data. By comparing magnesium's two outer electrons against sodium's one, they construct a mechanistic explanation for the reactivity difference. The teacher then anchors this to the Thika Road phenomenon (the street lamp's sodium vs. the welding torch's argon) and extends it to CAN fertiliser (Calcium Ammonium Nitrate) used by Rift Valley maize farmers, showing that Group 2 reactivity is engineered into the foods Kenyans eat every day.
Safety Notes	Dilute HCl (1 mol/L maximum) must be handled in a ventilated room or near an open window. Students must wear safety goggles and lab coats/aprons. Magnesium ribbon burns with intense UV light — DO NOT look directly at burning Mg. Lighted splint test for hydrogen must be performed by the teacher as a demonstration only. Hot water/steam station must be supervised; use heat-resistant gloves. First aid kit must be accessible. Dispose of reaction mixtures by neutralising with sodium hydrogen carbonate solution before pouring down the drain.

B. LESSON OVERVIEW

Students arrive knowing that sodium (Group 1, 1 outer electron) reacts violently with water. Today they ask: what happens with magnesium (Group 2, 2 outer electrons)? In a supervised experiment they test magnesium ribbon with (i) dilute HCl, (ii) cold water, and (iii) hot water/steam, recording observations and writing balanced equations. A structured class discussion surfaces the pattern — two outer electrons make Group 2 metals less reactive than Group 1. Students then connect this to Thika Road (sodium street lamps vs. argon-shielded welding) and to CAN fertiliser (calcium and ammonium nitrate) used by maize farmers in Nakuru and Uasin Gishu counties. The lesson closes with DQB updates and revision of element-sorting cards to include Group 2 evidence.

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**
Phase 1 — Predict 📹 **VIDEO:** Worked example: Finding the formula of an ionic compound Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **PDF:** Using group work in your classroom Source: TESSA - Teacher Resources 🔍 Search ARES for similar readings	Students receive a 'Predict Card' showing three unlabelled test tubes: (A) sodium in water — a sketch of vigorous fizzing and flame from Lesson 3; (B) magnesium ribbon in cold water — blank; (C) magnesium ribbon in dilute HCl — blank. Working individually for 3 minutes, they write a prediction for B and C using the sentence stems: 'I predict that magnesium in cold water will… because…' and 'I predict that magnesium in dilute HCl will… because…'. They then share with a shoulder partner for 2 minutes.	Project the three test-tube sketches on the board. Say: 'You have already seen what sodium — one outer electron, Group 1 — does to water. Magnesium sits one group to the right with TWO outer electrons. Before we touch any apparatus, write your prediction privately.' Allow 3 minutes of silent individual thinking (WAIT TIME). Then: 'Turn to your shoulder partner. You have 2 minutes — convince each other.' After 2 minutes, cold-call 3 students using name sticks: ask each to read their prediction aloud and state the reason. Record the three predictions verbatim on the board under the heading 'OUR PREDICTIONS — we will test these today.' Do NOT confirm or deny any prediction. Say: 'Keep these predictions visible — we are scientists checking our own thinking.'	Prediction Anchoring: by committing predictions to writing before evidence is gathered, students activate prior knowledge about outer electrons and reactivity, creating a cognitive gap that motivates careful observation. Predictions posted publicly create accountability and a reference point for revision.	Observable indicator: every student has written a prediction that references outer electrons OR reactivity OR a comparison to sodium. Teacher circulates during individual writing and notes any student who writes a blank or a guess with no reasoning — these students are prioritised for cold-calling and targeted questioning during the Explain phase.
Phase 2 — Observe 📹 **VIDEO:** Worked example: Finding the formula of an ionic compound Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **PDF:** Using group work in your classroom Source: TESSA - Teacher Resources 🔍 Search ARES	Station-based supervised experiment (groups of 4). Each group receives: 3 cm of magnesium ribbon (cleaned with sandpaper), a test tube of cold water with 2 drops of universal indicator, a test tube of dilute HCl (1 mol/L), a boiling tube of near-boiling water (teacher-prepared, steam rising). Students follow a printed procedure card: (1) Place Mg ribbon in cold water — observe for 5 minutes, record colour of indicator; (2) Place a fresh piece of Mg in dilute HCl — observe immediately, hold a damp litmus paper over the mouth of the tube (teacher demonstrates the lighted	Before experiment: spend 5 minutes on a mandatory safety briefing. Say verbatim: 'Goggles on before you touch anything. If HCl spills on skin, rinse immediately with cold water for 10 minutes. The lighted splint test is MY job today — do not attempt it.' Distribute apparatus only after every student has goggles on. Circulate continuously during the 15-minute practical. At the 5-minute mark, pause all groups: 'What colour is your universal indicator in the cold water tube? Call it out — group by group.' (This is a quick whole-class formative check.) At the 10-minute mark, pause again: 'What do you	Structured Observation with a Results Table: the pre-formatted table (reagents | observations | gas | equation slot) scaffolds scientific recording and prevents students from writing vague notes. The public whole-class results table on the board allows groups to compare observations in real time, surfacing discrepancies that drive discussion in the Explain phase. This enacts the NGSS Science and Engineering Practice: Planning and Carrying Out Investigations.	Observable indicator: each group's results table has at least one recorded observation per reaction (not blank). Teacher specifically checks that students have noted the DIFFERENCE in vigour between cold water and HCl reactions. Any group whose cold-water observation is 'nothing happened' is asked: 'Look at the ribbon surface under the light — can you see any tiny bubbles or a white film?' This probes for the subtle $Mg(OH)_2$ film observation.

for similar readings	splint test on the bench as a class demonstration only); (3) Teacher demonstration: Mg ribbon held in tongs over the steam from boiling water — students observe from their seats and record. All observations are entered into a structured results table (columns: reagents | observations | colour/pH change | gas produced | equation slot).	SEE in the HCl tube? I need one observation word from each group — no interpretations yet.' For the steam demonstration, say: 'Watch the ribbon surface. Watch the water level. Watch for any colour change in the indicator I am dropping into the steam condensate.' Allow 10 seconds of silent observation after each demonstration step (WAIT TIME). Record all group observations on the class results table projected on the board.		
Phase 3 — Explain 📹 **VIDEO:** Worked example: Finding the formula of an ionic compound Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **PDF:** Using group work in your classroom Source: TESSA - Teacher Resources 🔍 Search ARES for similar readings	Students return to seats. Using observations from Phase 2, they complete the equation slots in their results table: $Mg(s) + 2HCl(aq) \rightarrow MgCl_2(aq) + H_2(g)$ and $Mg(s) + H_2O(steam) \rightarrow MgO(s) + H_2(g)$. The teacher guides equation balancing using the board. Students then complete a 'Reactivity Comparison Table' contrasting sodium (Group 1) and magnesium (Group 2) across four columns: number of outer electrons | reaction with cold water | reaction with dilute acid | reaction with steam. Class discussion surfaces the mechanistic explanation: two outer electrons are held more tightly by the nucleus than one, making it harder for Mg to lose both electrons simultaneously, hence slower/less vigorous reactions. Teacher then connects to the Kenyan context: 'CAN fertiliser — Calcium Ammonium Nitrate — which Rift Valley maize and wheat farmers buy at Nakuru Farmers' Choice agro-dealers — contains calcium, another Group 2 element, two outer electrons. It dissolves steadily in soil water rather than explosively, releasing Ca^{2+} ions	Begin equation work: project a blank equation frame on the board. Say: 'Use your observations to fill in the products. You have 4 minutes — work alone first.' (WAIT TIME: 4 minutes silent work.) Cold-call 2 students to write their equations on the board. Ask the class: 'Does anyone want to challenge or refine what is written?' Allow 30 seconds of thinking (WAIT TIME). Then explicitly balance the equations step by step, narrating each step aloud. For the comparison table, say: 'Fill in the sodium row using what you know from Lesson 3. Fill in the magnesium row using today's data.' Cold-call 3 different students (total cold-call count for this phase: 5 students). For the CAN fertiliser connection, hold up a photograph of a CAN fertiliser bag (or draw the label on the board showing '$Ca(NO_3)_2 \cdot NH_4NO_3$'). Ask: 'Why would a Rift Valley farmer WANT a Group 2 metal in their fertiliser rather than a Group 1 metal?' Allow 15 seconds WAIT TIME. Cold-call 2 students. Confirm and extend their answers. Finally, return to the predictions posted in Phase 1: 'Let us read	Claim-Evidence-Reasoning (CER) Framework: students must state a claim ('Magnesium is less reactive than sodium'), cite specific evidence ('Mg produced slow bubbles in cold water; Na ignited immediately'), and provide reasoning ('because Mg has 2 outer electrons held more tightly than Na's 1 outer electron'). The Reactivity Comparison Table scaffolds the evidence column. The CAN fertiliser discussion applies CER to a real-world Kenyan context, reinforcing that chemical family properties have tangible agricultural consequences.	Observable indicator: student equations are balanced (correct coefficients, correct state symbols). Teacher specifically checks that students have written MgO (not $Mg(OH)_2$) as the steam product — a common error. The Reactivity Comparison Table is collected at the end of this phase as an exit artefact; teacher scans for students who have left the 'reasoning' column blank or have reversed the reactivity order (claiming Mg is MORE reactive). These students receive targeted follow-up in the next lesson.

	that improve soil structure and supply nutrients to ugali maize. If calcium behaved like sodium, it would blow up the shamba.' Students annotate their element-sorting cards (introduced in Lesson 1) to add: 'Group 2 — 2 outer electrons — LESS reactive than Group 1.'	Prediction 1 again. Was it supported? Partially? Refuted? How do you know?' Students annotate their Predict Cards with 'SUPPORTED / PARTIALLY SUPPORTED / REFUTED' in a different colour pen.		
Phase 4 — Driving Question Board (DQB) Update 📹 **VIDEO:** [Worked example: Finding the formula of an ionic compound] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **PDF:** [Using group work in your classroom] Source: TESSA - Teacher Resources 🔍 Search ARES for similar readings	Each student receives two sticky notes. On the BLUE note they write one new thing they can now explain about the anchoring phenomenon: 'I can now explain why… [connection to Thika Road / welding / CAN fertiliser / water-treatment tablet].' On the YELLOW note they write one question that today's evidence has raised or left unanswered. Students post notes on the class DQB under the columns 'WE CAN NOW EXPLAIN' and 'WE STILL WONDER'. Groups rotate around the DQB for 3 minutes to read peers' notes and place a dot sticker on any yellow note that matches their own lingering question (dot voting to identify the most pressing unresolved questions).	Say: 'The DQB tracks our growing understanding of the driving question: why do some elements explode in water, some glow in glass tubes, and others rust on a Kenyan farm? Today we added Group 2 evidence. Write what you can NOW explain that you could NOT explain at the start of this unit.' Allow 4 minutes for individual sticky-note writing (WAIT TIME). During the gallery walk, do NOT direct students — let them read and dot-vote independently. After the walk, read aloud the top 3 most dot-voted yellow questions. Say: 'These are the questions we carry forward. Let us add them to the top of our board.' Explicitly connect to the upcoming lessons: 'Lessons 7 and 8 will bring in transition metals — iron, copper, zinc — which will help us answer the question several of you have raised about the mabati roof.' Record the three questions on a dedicated 'Questions Carried Forward' section of the DQB.	Public Knowledge Tracking (DQB): the Driving Question Board makes the class's collective understanding visible and cumulative. By distinguishing 'can explain' from 'still wonder', students self-regulate their own understanding gaps. Dot voting on unresolved questions enacts democratic scientific prioritisation and mirrors how research communities identify knowledge frontiers.	Observable indicator: blue sticky notes contain a specific reference to the phenomenon (not a generic science statement). Yellow sticky notes are questions, not statements. Teacher reads all blue notes before Lesson 7 to identify misconceptions — particularly any student who writes that 'Group 2 is more reactive than Group 1' — and plans a corrective hook for the next lesson's predict phase.
Phase 5 — Model Building 📹 **VIDEO:** [Worked example: Finding the formula of an ionic compound] Source: Khan	Students retrieve their 'Element Behaviour Model' — a visual diagram begun in Lesson 1 showing elements grouped by family with arrows labelled with reactivity behaviours. Today they add a Group 2 column alongside their existing Group 1 column. They draw: (a) a Mg atom	Distribute students' model folders. Say: 'Your model must now show WHY Group 2 behaves differently from Group 1. Draw the electron configuration. Draw the direction of electron loss. Write the comparison.' Allow 8 minutes of individual model work. Circulate and ask probing questions to 4–5	Cumulative Diagrammatic Modelling: by physically adding to the same visual model across all 8 lessons, students construct a coherent explanatory framework rather than isolated lesson notes. The requirement to annotate the original phenomenon sketch ensures every model addition is	Observable indicator: the updated model shows (i) Mg electron configuration 2,8,2 correctly drawn, (ii) a comparative annotation referencing Group 1 vs. Group 2 outer electrons, (iii) at least one Kenyan context label. Teacher stamps or ticks models that meet all three criteria. Models

Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **PDF:** Using group work in your classroom Source: TESSA - Teacher Resources 🔍 Search ARES for similar readings	(electron configuration 2,8,2) showing 2 outer electrons; (b) two arrows leaving the outer shell labelled 'lost to form Mg^{2+}'; (c) a comparative size annotation: 'nucleus pulls 2 electrons harder than Na's nucleus pulls 1 — harder to lose, less reactive.' They also add a Kenyan context box: 'CAN fertiliser (Nakuru/Rift Valley) contains Ca^{2+} — Group 2, controlled reactivity, safe for soil.' Finally, students annotate the original phenomenon sketch (Thika Road street lamp vs. jua kali welder) with a new label: 'Na in lamp = Group 1, 1 outer e^-, reacts to give orange light; Mg/Ca in fertiliser = Group 2, 2 outer e^-, reacts slowly, nourishes crops.'	students: 'Why does your arrow go OUTWARD from the outer shell?' and 'What would happen to this model if we moved one column further right to Group 3?' (anticipating transition metals). With 2 minutes remaining, ask 2 students to share their models under the document camera (cold-call using name sticks). Prompt the class: 'What has this student done well? What could be added?' Close by saying: 'By Lesson 8, your model must explain ALL six families — alkali metals, alkaline earth metals, halogens, noble gases, transition metals, and the specific Kenyan contexts for each. Today you have completed two of the six. Keep your model — it becomes part of your unit portfolio.'	explicitly tied to the anchoring context, preventing abstract memorisation. This enacts the NGSS Science and Engineering Practice: Developing and Using Models.	missing the comparative annotation are flagged — these students will be paired with a peer model-mentor at the start of Lesson 7.

D. TEACHER REFLECTION

After the lesson, reflect on the following questions and note responses in your professional journal:

1. SAFETY EXECUTION: Were all students wearing goggles throughout the practical? Did any group attempt the lighted splint test unsupervised? What adjustments will you make to station management in Lesson 7 when students investigate transition metal properties?
2. EQUATION BALANCING: How many students correctly balanced $Mg(s) + H_2O(steam) \rightarrow MgO(s) + H_2(g)$ on the first attempt? If fewer than 60% succeeded, plan a 5-minute targeted review at the start of Lesson 7 using a partially completed equation frame.
3. REACTIVITY COMPARISON: Did students articulate the mechanistic reason (nuclear attraction on 2 outer electrons vs. 1) or did they only state the observation ('Mg reacted slower')? Students who only stated observations need the electron configuration model explicitly re-connected to reactivity in the model-building phase of Lesson 7.
4. KENYAN CONTEXT CONNECTION: Were students able to explain WHY a farmer in Uasin Gishu would prefer CAN over a Group 1-based fertiliser? If this connection was weak, source a CAN fertiliser bag or photograph for Lesson 7's opening hook.
5. DQB QUALITY: Review the yellow sticky notes. Are the 'still wonder' questions pointing toward transition metals, bonding, or reactivity series — which are exactly the content of Lessons 7 and 8? If students' questions are unrelated to the driving question, the DQB protocol may need more explicit scaffolding next time.
6. EQUITY AUDIT: Were all four group members actively recording data, or were one or two students passive? Reassign roles for Lesson 7 so that students who were passive observers today are the designated 'chief recorder' and 'equation writer.'

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Magnesium ribbon placed in cold water produced very slow, tiny bubbles and a faint white film ($Mg(OH)_2$) — the universal indicator

	barely changed. Magnesium in dilute HCl produced rapid, vigorous bubbling; the gas tested positive for hydrogen (squeaky pop). Magnesium ribbon held over steam reacted moderately, producing a white solid (MgO) and hydrogen gas. In all three cases, the reaction was noticeably slower and less dramatic than the sodium-in-water demonstration from Lesson 3.
What did I learn?	Magnesium (Group 2, electron configuration 2,8,2) has two outer electrons. Both must be lost to form the Mg^{2+} ion. Because the nucleus attracts two outer electrons more effectively than sodium's nucleus attracts its single outer electron, it is harder for magnesium to lose electrons — making Group 2 elements less reactive than Group 1 elements. Balanced equations: $Mg(s) + 2HCl(aq) \rightarrow MgCl_2(aq) + H_2(g)$; $Mg(s) + H_2O(steam) \rightarrow MgO(s) + H_2(g)$. Calcium (also Group 2) is the active ingredient in CAN fertiliser (Calcium Ammonium Nitrate) used by Rift Valley maize and wheat farmers — its controlled reactivity means it releases Ca^{2+} ions steadily into the soil rather than reacting explosively.
How does this explain the phenomenon?	The anchoring phenomenon asks why some elements react violently and others do not. Today's evidence adds a new layer: it is not just WHICH group an element is in, but HOW MANY outer electrons it has and how tightly the nucleus holds them. Sodium (1 outer electron, Group 1) loses it easily → violent reaction with water → orange glow in a Thika Road street lamp. Magnesium (2 outer electrons, Group 2) loses them more reluctantly → slower, controlled reactivity → safe for agricultural use in CAN fertiliser on Kenyan farms. The same periodic table position that makes sodium spectacular makes calcium useful — chemical family properties are engineered into the materials that sustain Kenyan daily life.

LESSON 7 (80 minutes (2 × 40-minute lessons)): Transition Metals: The Workers of the Periodic Table — Iron, Copper, and Zinc in Kenyan Life	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To investigate the physical and chemical properties of selected transition metals (iron, copper, zinc) and explain how those family properties make them indispensable in Kenyan construction, agriculture, and electrical infrastructure.
Knowledge	Learners will be able to: (a) identify transition metals as a chemical family occupying the central block of the periodic table; (b) describe the characteristic physical properties of transition metals (high melting points, hardness, malleability, electrical and thermal conductivity, variable oxidation states, formation of coloured compounds); (c) describe selected chemical properties including reaction with oxygen, dilute acids, and formation of oxides; (d) outline the uses of iron, copper, and zinc and their compounds in day-to-day Kenyan life.
Skills	Learners will be able to: (a) carry out activities to investigate and compare physical properties (lustre, conductivity, malleability) of iron, copper, and zinc samples; (b) observe and record the reaction of copper and zinc with dilute hydrochloric acid and compare with iron; (c) construct and annotate a property–use evidence table; (d) write balanced chemical equations for the reactions of transition metals with oxygen and dilute acids; (e) update the class Driving Question Board and cumulative model with transition metal evidence.
Attitudes	Learners will: (a) appreciate how the specific properties of transition metals underpin the built environment and agricultural tools of Kenya (jembe, mabati roof, electrical wiring, galvanised steel water tanks); (b) observe the value — Love — by taking care of self and others by following safety procedures when handling dilute acids and metal samples; (c) develop Responsibility by using laboratory apparatus and digital resources carefully.
Key Inquiry Question	How are elements suited to their use in day-to-day life?
Purpose in Storyline	Lessons 1–6 established the reactive alkali metals (sodium), the halogens (chlorine), and the noble gases (argon, helium) — explaining three corners of the anchoring phenomenon. Lesson 7 now tackles the fourth corner and beyond: why is the jembe (hoe) made of iron, the electrical wiring of copper, and the mabati roof coated in zinc? Students discover that transition metals form a distinct chemical family whose high melting points, hardness, conductivity, and resistance to rapid corrosion make them the 'worker' elements — contrasting sharply with the explosive alkali metals and inert noble gases already studied. This lesson is the penultimate evidence lesson before the final synthesis in Lesson 8, so learners must consolidate all four families onto one coherent model.
Safety Notes	1. Dilute hydrochloric acid (1 mol/L maximum): learners must wear safety goggles and avoid skin contact; teacher demonstrates acid dispensing before learners handle it. 2. Metal filings/turnings: handle with tongs or spatulas — no bare hands (zinc dust is irritant). 3. Heating copper wire in a flame: use tongs; allow to cool on a ceramic tile before touching. 4. First-aid kit must be visible and accessible. 5. No concentrated acids. 6. Wash hands thoroughly after the practical.

B. LESSON OVERVIEW

Learners begin by predicting whether a jembe (iron hoe), a copper electrical wire, and a zinc-coated mabati roof panel would behave like sodium (explodes in water) or argon (does nothing) — connecting directly to the anchoring phenomenon. They then carry out a hands-on investigation comparing the physical properties (lustre, malleability, conductivity) and chemical reactivity (with dilute HCl) of iron, copper, and zinc samples, recording evidence in a structured Property–Use table. An annotated diagram of a Kenyan hardware shop window (jua kali tools, wiring, roofing sheets) serves as a contextual reading that links each property to a real use. Learners write and balance chemical equations for the reactions observed, then use a 'Family Portrait' sensemaking strategy to compare all four families studied so far. The lesson closes with a DQB update and a cumulative model revision that now incorporates all four chemical families, positioning learners to write the full explanatory

narrative in Lesson 8.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Phase 1 — PREDICT (10 minutes) 📹 **VIDEO:** Video: pH Scale Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** Transition Metals (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Each learner receives a 'Mystery Metals' card showing three images: (1) a jembe (iron hoe) rusting on a Rift Valley farm, (2) bare copper wiring inside a Nairobi flat under construction, (3) a shiny zinc-coated mabati roof in Kibera. Learners individually write answers to two prediction questions in their science journals: (Q1) 'These three materials are all transition metals — pure elements. Predict whether each will react violently, slowly, or not at all with water and with a dilute acid. Give a reason using evidence from Lessons 1–6.' (Q2) 'Sodium is also a metal and an element. Predict at least TWO ways in which iron, copper, and zinc will behave DIFFERENTLY from sodium. Explain why you think so.' Learners share predictions with a seat partner (Think-Pair) for 2 minutes.	Distribute Mystery Metals cards face-down. Say: 'Before you flip the card, think back to the street lamp sodium from Lesson 2 — what happened when sodium touched water? [WAIT 10 seconds] Good. Now flip the card.' Read Q1 and Q2 aloud. Circulate for 4 minutes of silent individual writing — do NOT answer questions; redirect with 'What do you already know about metals from Lessons 1–6?' After Think-Pair, cold-call 3 learners (choose one who predicted 'violent', one 'slow', one 'no reaction') and record their predictions verbatim on the board under the heading PREDICTED. Say: 'We will test every prediction today. Keep your journals open — you will revise these predictions at the end of the lesson.'	Think-Pair-Predict: activates prior knowledge from Lessons 1–6 (sodium reactivity, noble gas inertness) and creates cognitive dissonance — learners who predicted 'violent like sodium' will be surprised; learners who predicted 'inert like argon' will also be surprised. The prediction anchors all subsequent observation to the same question, sustaining the driving question thread.	Teacher scans journals during circulation — observable indicator: learner has written a prediction AND cited at least one piece of evidence from a previous lesson (e.g., 'sodium reacted violently because it is in Group 1'). Learners who cite no prior evidence are prompted: 'What did we observe in Lesson 2 that could help you here?'
Phase 2 — OBSERVE (30 minutes) 📹 **VIDEO:** Video: pH Scale Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** Transition Metals (Flexbook) Source: CK-12 🔍 Search ARES	Groups of 4 carry out a two-part structured investigation. PART A — Physical Properties (15 min): Each group receives labelled samples of iron (steel wool / iron nail), copper (short length of electrical wire), and zinc (zinc granule or a small piece of galvanised sheet). Using the Property–Use Evidence Table (pre-printed in their workbooks), learners: (i) observe and record lustre (shiny/dull) and colour; (ii) test malleability by gently pressing each sample with a spatula handle	Before Part A: 'You are going to compare three transition metals. Your job is to gather EVIDENCE — exact observations, not guesses. Write what you SEE, HEAR, or FEEL (carefully!). I will give you 15 minutes.' Set a visible timer. Circulate — at 8 minutes, pause the class: 'Table 2, what did you observe for copper conductivity? [Cold-call 1 learner — WAIT 10 seconds] Good — does everyone agree? Thumbs up/down.' Before Part B: Demonstrate acid dispensing at	Structured Evidence Table: the pre-formatted Property–Use Evidence Table scaffolds observation into comparable columns (Property → Observed Result → Kenyan Use Example → Reason the Property Matters). This is an NGSS Science and Engineering Practice 3 (Planning and Carrying Out Investigations) and Practice 4 (Analysing and Interpreting Data). By placing melting point data from the reference card alongside their own conductivity and malleability	Observable indicator for Part A: completed rows for all three metals in the Property–Use Evidence Table with at least 3 properties recorded per metal. Observable indicator for Part B: a written word equation present for iron + HCl and zinc + HCl (even if unbalanced); 'no visible reaction' recorded for copper + HCl. Teacher checks 4 tables during Part B — if word equations are absent, prompt: 'What two substances went IN? What did you observe coming OUT?'

for similar readings	against a ceramic tile — record 'bends / does not bend / flakes'; (iii) test electrical conductivity using a simple circuit card (battery, bulb, two crocodile-clip leads) — touch leads to each metal and record 'bulb lights / does not light'; (iv) record melting point data from a reference card provided (Fe: 1538 °C; Cu: 1085 °C; Zn: 420 °C — compare to Na: 98 °C and Cl_2: −101 °C). PART B — Chemical Reactivity with Dilute HCl (15 min): Teacher first demonstrates the acid dispensing procedure (3 cm^3 into a test tube using a dropper — goggles on). Groups then place a small iron nail, a copper piece, and a zinc granule into separate labelled test tubes and add 3 cm^3 of 1 mol/L HCl to each. Learners observe for 3 minutes, recording: bubble production (vigorous / slow / none), colour change of solution, any heat produced (touch outside of tube gently). They write word equations for any reactions they observe. Groups also view a 30-second video clip (pre-loaded on the ARES offline tablet) showing copper wire being heated in a Bunsen flame — observing the black copper(II) oxide layer that forms, contrasting with the non-reactive argon gas used in the welding torch from the phenomenon.	the front bench. Say explicitly: 'Goggles on NOW before I place acid on any bench. I am watching.' Dispense acid to groups only after confirming all goggles are on. During Part B, cold-call 2 groups at 5 minutes: 'What do you observe in the zinc tube? [WAIT 12 seconds] Use the word EVIDENCE in your answer.' After the video clip: 'Connect what you just saw — copper turning black — to our phenomenon. Which substance in our phenomenon also changes on contact with something in the environment? [WAIT 15 seconds] Discuss with your group for 1 minute, then I will cold-call.' Cold-call 2 groups.	observations, learners build a multi-property evidence base rather than a single isolated data point — directly supporting the later 'Family Portrait' sensemaking.	
Phase 3 — EXPLAIN (20 minutes) 📹 **VIDEO:** Video: pH Scale Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos	STEP 1 — Equation Writing (8 min): Learners use their observations to write and balance symbol equations for: (a) iron + hydrochloric acid → iron(II) chloride + hydrogen; $Fe + 2HCl \rightarrow FeCl_2 + H_2$; (b) zinc + hydrochloric acid → zinc chloride + hydrogen; $Zn + 2HCl \rightarrow ZnCl_2 + H_2$; (c) copper + oxygen (on heating) →	For equation writing: write Fe + HCl → ? on the board and say: 'Your observations told you gas was produced. What gas? [WAIT 12 seconds] How do we show that in a symbol equation? Work in pairs for 3 minutes, then I will cold-call.' Cold-call 2 pairs — one for iron, one for zinc. For copper oxide: 'You watched copper turn	Family Portrait Poster: an NGSS Practice 6 (Constructing Explanations) strategy that requires learners to synthesise evidence from seven lessons into a single comparative framework. The four-column structure makes the pattern of family properties visible as a spatial pattern — mirroring the logic of the periodic	Observable indicator: each group's Family Portrait has all four columns populated with at least one Kenyan example per column AND at least one chemical equation or chemical behaviour descriptor per column. Teacher specifically checks whether the Transition Metals column contrasts with Alkali Metals on reactivity — if

📖 **HTML:** [Transition Metals (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	copper(II) oxide; $2Cu + O_2 \rightarrow 2CuO$. Learners also write the equation for iron rusting (simplified): $4Fe + 3O_2 \rightarrow 2Fe_2O_3$, connecting to the rusting jembe image from the Mystery Metals card. STEP 2 — Family Portrait (10 min): Each group receives a blank 'Family Portrait' poster template with four columns: Alkali Metals | Halogens | Noble Gases | Transition Metals. Using sticky notes (or writing directly), groups populate each column with: (i) one Kenyan example element, (ii) two key physical properties, (iii) one key chemical behaviour, (iv) one Kenyan use. Groups draw on evidence from all seven lessons. STEP 3 — Gallery and Cold-Call (5 min): Two groups share their Family Portrait columns for Transition Metals aloud. Class votes (thumbs) on accuracy. Teacher annotates the class master poster on the board. Learners add a 'Transition Metals' row to their running Chemical Families summary table in their journals.	black. What family does oxygen belong to? [WAIT 10 seconds] Write the equation for copper reacting with oxygen — you have 2 minutes.' Cold-call 1 learner to write on the board; class checks by atom-counting aloud together. For rusting: 'The jembe on our Mystery Metals card is rusting slowly — not explosively like sodium. What does that tell us about transition metal reactivity compared to Group 1? [WAIT 15 seconds] Discuss with your partner — 1 minute.' Cold-call 1 learner. For Family Portrait: 'You have evidence from Lessons 1 to 7. Fill in all four columns — you have 8 minutes. Every group member must write at least one sticky note.' Circulate and challenge incomplete entries: 'What evidence supports that claim?' During gallery: 'Listen carefully — when Group 3 shares, you are checking their evidence against YOUR data. Thumbs up if you agree, thumbs sideways if you are unsure.'	table itself. The Kenyan use column maintains the phenomenon thread: every property claim must be linked to a real Nairobi/Rift Valley/Mombasa context, preventing abstract memorisation.	a group writes 'both react violently with water', the teacher prompts: 'What did you observe in Part B today that tells you about transition metal reactivity with acid? Does iron behave like sodium in water?'
Phase 4 — DRIVING QUESTION BOARD UPDATE (8 minutes) 📹 **VIDEO:** [Video: pH Scale] Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** [Transition Metals (Flexbook)]	Learners individually write two items on sticky notes: (1) a GREEN 'We Now Know' note — one sentence summarising what today's evidence adds to the explanation of the phenomenon (e.g., 'Transition metals like iron, copper, and zinc are hard, have high melting points, and react slowly or not at all with water — that is why they are used for jembes, wiring, and mabati roofs, unlike sodium which explodes'); (2) an ORANGE 'Still Wondering' note — one question that today's lesson raised or left unanswered (e.g., 'Why does zinc protect iron from	Say: 'Take two sticky notes — one green, one orange. You have 3 minutes to write. Your green note must mention at least ONE transition metal property and ONE Kenyan object from today.' Set timer. After 3 minutes: 'Come and place your notes.' Once all notes are placed, read 3 orange notes aloud (choose ones that connect to galvanisation / alloys / variable oxidation states). Say: 'These are excellent scientific questions. The question about zinc protecting iron is called GALVANISATION — we will resolve it in Lesson 8 when we build our final model. Write the	Driving Question Board (DQB) Update: an NGSS Practice 1 (Asking Questions) and Practice 8 (Obtaining, Evaluating, and Communicating Information) strategy. The two-colour sticky-note protocol makes the distinction between established evidence and remaining uncertainty visible to the whole class. Over seven lessons the DQB has grown into a class-level epistemic artefact — learners can physically see how their explanation has expanded lesson by lesson, building metacognitive awareness of their own knowledge construction.	Observable indicator: every learner posts one green and one orange note. Teacher scans green notes — acceptable notes name a specific property AND a specific Kenyan object. Unacceptable green notes (e.g., 'transition metals are metals') are gently returned with the prompt: 'Connect this to something real in Kenya — which object and which property?'

Source: CK-12 🔍 Search ARES for similar readings	rusting on a mabati roof — what is actually happening chemically?'). Learners place notes on the class Driving Question Board in the correct columns. The class reads 3–4 'Still Wondering' notes aloud — teacher confirms which will be addressed in Lesson 8 (the synthesis lesson) and which extend beyond Grade 10.	word galvanisation in your journal as a preview.' Cold-call 1 learner: 'Using today's evidence, can you now explain why the jembe is made of iron and not sodium? [WAIT 15 seconds]' Record the learner's answer verbatim under 'Our Growing Explanation' on the DQB.		
Phase 5 — MODEL BUILDING (12 minutes) 📹 **VIDEO:** Video: pH Scale Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** Transition Metals (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners retrieve their cumulative Chemical Families Model — a diagram they have been building since Lesson 1, showing the anchoring phenomenon elements (Na, Ar, Cl, He) with annotated property arrows explaining each behaviour. Today they ADD a new section to the model: a 'Transition Metals Zone' containing Fe, Cu, and Zn, annotated with: (a) at least 3 physical properties (with measured/observed evidence from today's practical); (b) at least 1 chemical reaction shown as a balanced equation; (c) arrows connecting each element to its Kenyan use object (Fe → jembe, Cu → electrical wiring, Zn → mabati coating); (d) a CONTRAST arrow comparing transition metal reactivity to alkali metal reactivity (e.g., 'Fe reacts slowly with dilute HCl; Na reacts explosively with water — both are metals but different families'). Learners also add a 'Why are they different?' annotation box — in their own words explaining that position in the periodic table (central d-block vs Group 1 s-block) accounts for the behavioural difference. Learners who finish early begin drafting a 2–3 sentence 'Phenomenon Explanation So Far' paragraph at the bottom of their model.	Say: 'Open your cumulative models. You have been building this since Lesson 1 — today we add the transition metals. Your model must show EVIDENCE, not just labels. Use arrows, colours, and equations.' Set timer for 10 minutes. Circulate — at 5 minutes, pause the class: 'Check your model — does it show a CONTRAST between today's metals and the sodium from Lesson 2? If not, add a contrast arrow now.' Cold-call 1 learner at 9 minutes: 'Read me your contrast annotation — what is the most important difference you have captured? [WAIT 12 seconds]' Close the lesson: 'In Lesson 8, you will use this model to write a complete explanation of our phenomenon — every element, every family, every property. Make sure your model is complete and legible before next class. We will do a peer-review of models at the start of Lesson 8.' Assign home reflection: 'Tonight, look at one object in your home that is made of a transition metal. Write its name, the metal it contains, and one property from today that explains why that metal was chosen.'	Cumulative Model Revision (NGSS Practice 2 — Developing and Using Models): the running model functions as a knowledge-integration artefact. Each lesson's evidence is not stored in isolation but grafted onto a growing explanatory structure. By requiring CONTRAST arrows between families, the protocol enforces relational understanding — learners cannot simply add new facts; they must show how the new facts change the overall pattern. The 'Phenomenon Explanation So Far' paragraph bridges the model (visual) to the driving question (verbal), preparing for the Lesson 8 synthesis.	Observable indicator: the model now has a labelled Transition Metals Zone with at least 3 annotated properties, at least 1 balanced equation, at least 1 Kenyan use object, and at least 1 contrast arrow to another family. Exit check: teacher scans 6–8 models before learners leave — any model missing the contrast arrow receives the verbal prompt: 'Your model shows what transition metals DO — but does it show how they are DIFFERENT from Group 1? Add that before Lesson 8.'

D. TEACHER REFLECTION

After the lesson, consider the following questions:

1. PREDICTION QUALITY: Did learners' initial predictions show genuine use of prior evidence from Lessons 1–6, or were predictions generic ('metals react with acid')? If predictions were shallow, the connecting thread between lessons may need explicit reinforcement at the start of Lesson 8.

2. PRACTICAL SAFETY: Were all learners wearing goggles before acid was dispensed? If any group began without goggles, revisit the safety protocol at the start of the next lesson and consider assigning a 'Safety Monitor' role within each group.

3. EQUATION BALANCING: Were learners able to balance $Fe + 2HCl \rightarrow FeCl_2 + H_2$ independently, or did most require significant scaffolding? If fewer than 60% of learners produced a correctly balanced equation, build a 5-minute equation-balancing warm-up into the start of Lesson 8 before the synthesis activity.

4. FAMILY PORTRAIT DEPTH: Did learners' Family Portrait posters show genuine comparison across all four families, or did they treat transition metals in isolation? The quality of Lesson 8 synthesis depends on this comparative understanding — if Family Portraits were weak, consider a 10-minute structured class discussion at the start of Lesson 8 to consolidate the four-family comparison before model writing.

5. DQB ORANGE NOTES: What 'Still Wondering' questions emerged? Notes about galvanisation, alloys, or why copper is preferred over iron for wiring signal readiness for extension. Notes suggesting basic confusion (e.g., 'I still don't understand what a transition metal is') indicate the need to re-teach the defining characteristics before the Lesson 8 synthesis.

6. HOME REFLECTION TASK: Collect and scan the home reflection entries at the start of Lesson 8 — they provide a formative window into whether learners can transfer property–use reasoning to unseen objects beyond the classroom context.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Learners observed the physical properties (lustre, malleability, electrical conductivity, high melting points) and chemical reactivity (slow reaction of iron and zinc with dilute HCl; no reaction of copper with dilute HCl; copper turning black on heating in air) of iron, copper, and zinc samples. They also observed video evidence of copper oxidation in a flame, contrasting with the inert argon used in the welding torch from the anchoring phenomenon.
What did I learn?	Transition metals form a distinct chemical family with high melting points, hardness, malleability, and good electrical and thermal conductivity. Unlike alkali metals (sodium), they react slowly or not at all with cold water, and react slowly with dilute acids. They form coloured compounds (e.g., blue copper(II) sulphate, green iron(II) chloride) and have variable oxidation states. These properties directly explain why iron is used for jembes, copper for electrical wiring, and zinc for coating mabati roofing sheets in Kenya.
How does this explain the phenomenon?	The contrasting behaviours of sodium (Group 1 alkali metal — explosive reactivity) and iron/copper/zinc (transition metals — slow, controlled reactivity) are explained by their positions in the periodic table and the resulting differences in electron configuration and ionisation energy. Transition metals, sitting in the d-block, have higher nuclear charge and more complex electron structures that make them less reactive than Group 1 metals but still reactive enough to be chemically useful. This explains why the jembe (iron) rusts slowly over months rather than exploding instantly like sodium, and why copper wiring conducts electricity reliably without reacting with air under normal conditions — linking directly to the anchoring phenomenon of the Thika Road street lamp (sodium), the welding torch (argon), and extending the explanation to include the 'silent worker' metals of Kenyan daily life.

LESSON 8 (80 minutes): Final Explanation: Why Do Chemical Families Determine How Elements Build, Light Up, and Sustain Our Daily Lives?	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To consolidate and synthesise understanding of all chemical families — alkali metals, alkaline earth metals, halogens, noble gases, and transition metals — by constructing a final explanatory model that accounts for all phenomena observed across the unit, from the sodium street lamps of Thika Road to the bleaching of a hibiscus flower with chlorine water.
Knowledge	Students will be able to: (a) identify chemical families in the periodic table; (b) classify elements of the periodic table into their chemical families; (c) describe the physical and chemical properties of selected elements; (d) outline the uses of elements and their compounds in day-to-day life; (e) explain how periodic trends (reactivity, electronegativity, atomic radius) within each family account for the observed properties of elements; (f) write and balance equations for key reactions (Na + H_2O, Cl_2 + H_2O, Cl_2 + NaOH, Fe rusting, noble gas discharge in sealed tubes).
Skills	Constructing and comparing explanatory models; writing balanced chemical equations; evaluating evidence against predictions; synthesising information from multiple lessons into a coherent explanatory framework; presenting scientific arguments with supporting evidence.
Attitudes	Appreciate the uses of elements of the periodic table in day-to-day life; value the systematic organisation of the periodic table as a powerful predictive tool; demonstrate curiosity about how electron arrangement drives the behaviour of every element encountered in Kenyan daily life.
Key Inquiry Question	How are elements suited to their use in day-to-day life?
Purpose in Storyline	This is the capstone lesson of the unit. Students have gathered evidence across seven lessons — from alkali metal reactions with water, to the glow of noble gases in discharge tubes, to the bleaching of a hibiscus flower with chlorine water, and iodine's starch test on maize ugali flour. In this Final Explanation lesson, students compare their Lesson 1 naive model with their fully revised model, complete all open DQB questions, and produce a written Final Explanation that answers the anchoring phenomenon: why do sodium glow in a Thika Road street lamp, argon shield a jua kali weld, chlorine compounds purify Nairobi Water, and helium float harmlessly in a party balloon — while iron, copper, and zinc do something entirely different? All behaviours are now explained through the lens of chemical family membership and electron configuration.
Safety Notes	No new hazardous chemicals are introduced in this lesson. If chlorine water from Lesson 7 is referenced or displayed, ensure the container remains sealed. Iodine solution should be kept in a dropper bottle; avoid skin/eye contact. Remind students not to taste any chemicals. Wash hands after any contact with iodine. The teacher demonstration involving iodine and ugali flour paste should be conducted at the front with students observing from their seats.

B. LESSON OVERVIEW

Lesson 8 is the Final Explanation capstone for Sub-Strand 2.2. Students open by revisiting their very first Lesson 1 prediction models and the original anchoring phenomenon (Thika Road street lamps / jua kali welding / chlorine water tablets / helium balloon). Working in home groups, they compare their naive L1 models with their fully evolved L8 models, identify every conceptual upgrade, and articulate what evidence from each lesson drove that upgrade. The teacher leads a whole-class Final Explanation discussion using a structured 'Claim–Evidence–Reasoning' (CER) framework, ensuring each chemical family (alkali metals, alkaline earth metals, halogens, noble gases, transition metals) is explicitly connected to at least one Kenyan daily-life example and one balanced equation. Students then complete the

Driving Question Board by answering all remaining sticky-note questions — including "What about elements that do NOTHING at all?" (noble gases) posted in Lesson 7. The lesson closes with students writing individual Final Explanation paragraphs that answer the unit driving question in full, followed by teacher-led synthesis and a forward bridge to Sub-Strand 2.3 (Chemical Bonding).

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Phase 1 — Predict: Revisiting the Naive Model (L1 vs L8) 📹 **VIDEO:** Naming ions and ionic compounds Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Chemical Bonding (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Students receive back their Lesson 1 prediction drawings/models (stored by the teacher) showing their initial ideas about why sodium glows, argon is inert, chlorine bleaches, and iron rusts. Working in pairs, they annotate their L1 model with the symbol '?' beside every claim they now know was incomplete or wrong, and '✓' beside any claim that turned out to be correct. They then write one sentence predicting: 'By the end of today, I will be able to explain ____ which I could not explain in Lesson 1.' Pairs share their sentence with their group of four before whole-class sharing of two or three responses via cold-call.	Distribute L1 models collected in Lesson 1 (or printed screenshots from the ARES platform). Say: 'Take 3 minutes — no talking — to read your own Lesson 1 model. Think about everything you have learned since then.' [WAIT 3 minutes, silent individual reading.] Then: 'In your pairs, spend 4 minutes annotating with ? and ✓. I will cold-call three pairs to share one ? and one ✓ each.' [WAIT 4 minutes.] Cold-call Pair A: 'Amina and Brian — share one ? from your L1 model.' [WAIT 10 seconds for response.] Cold-call Pair B: 'Kipchoge and Faith — share one claim you marked ✓.' [WAIT 10 seconds.] Cold-call Pair C: 'Zawadi and Otieno — read your prediction sentence for today.' [WAIT 10 seconds.] Record two or three prediction sentences on the board under the heading 'What we still need to explain today.' Do NOT correct or validate yet — preserve cognitive tension.	Strategy: Elicit–Prior Knowledge Activation via Model Annotation. By physically holding their L1 models and annotating them, students activate metacognitive awareness of their own learning trajectory. The '?' and '✓' symbols externalise conceptual change, making growth visible. This creates productive dissonance that motivates the Final Explanation phase.	Observable indicator: Each pair produces at least two '?' annotations and one '✓' annotation on their L1 model. Teacher scans the room during pair work; if a pair has zero '?' marks, prompt: 'Look at what you wrote about WHY sodium glows — do you think that explanation is complete now?' Listen for the quality of prediction sentences; students who can name a specific concept (e.g., 'electron configuration,' 'Group 17 trend') are demonstrating deeper consolidation than those who write vague generalisations.
Phase 2 — Observe: Evidence Gallery Walk — Eight Lessons in Review 📹 **VIDEO:** Naming ions and ionic compounds	Eight A3 evidence posters are displayed around the classroom walls, one per previous lesson (L1–L7, plus one 'Transition Metals' summary). Each poster contains: the supporting phenomenon from that lesson, the key observation, and a blank 'Explanation gap' box. Students do a timed Gallery Walk (12 minutes,	Set the Gallery Walk: 'You have 12 minutes. Move as a group. Each person writes ONE explanation sentence per poster — your own words, not copied from a neighbour. Go.' [Set a visible timer. Circulate continuously.] At Poster 3 (sodium + water), listen for students using 'Group 1,' 'one valence electron,' or 'highly	Strategy: Evidence Gallery Walk with Explanation Gap Fill. This strategy requires students to retrieve, apply, and articulate — not merely recognise — explanations for each phenomenon. By writing individual sentences at each poster, every student produces their own explanatory language rather than	Observable indicator: Each student produces at least six sticky-note sentences across the eight posters (some posters will have multiple visits; groups may run out of time on the last one or two). Teacher reads sticky notes during the walk; flag any that are purely descriptive ('sodium reacted with water') rather than

Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Chemical Bonding (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	~90 seconds per poster), working in their home groups of four. Each student carries a sticky note. At each poster they read the phenomenon + observation, then write on their sticky note ONE sentence that fills the 'Explanation gap' box using the chemical family concept. They stick their sentence on the poster. Examples of poster prompts: Poster 3 — 'Sodium dropped into a trough of water at Thika Technical: fizzing, heat, violet flame. Explanation gap: ___'; Poster 6 — 'Hibiscus flower placed in chlorine water: petals bleach white within 2 minutes. Explanation gap: ___'; Poster 7 — 'Iodine solution dropped onto ugali flour paste: instant blue-black colour. Explanation gap: ___'.	reactive.' If a student writes only 'sodium reacts because it is an alkali metal,' prompt: 'Good start — can you add WHY alkali metals react that way? Think electron configuration.' [WAIT 10 seconds.] At Poster 6 (hibiscus bleaching), if students write 'chlorine bleaches because it is a halogen,' prompt: 'What does chlorine DO chemically to cause bleaching? Think about the equation $Cl_2 + H_2O$.' [WAIT 10 seconds.] At Poster 5 (noble gas discharge tube), if students write 'noble gases glow because they have full shells,' push: 'Full outer shells explain why they are INERT — but what causes the GLOW? Think about what happens when electricity passes through the gas.' [WAIT 10 seconds.] With 2 minutes left, call: 'Finish your current poster and return to your seats.'	free-riding on group consensus. The spatial distribution of posters across the room also prevents groups from clustering and copying.	explanatory ('sodium reacted vigorously because its single valence electron is easily lost, releasing hydrogen gas and forming NaOH'). These flagged students are prioritised for cold-calling in Phase 3.
Phase 3 — Explain: Final Explanation — Claim, Evidence, Reasoning (CER) 📹 **VIDEO:** Naming ions and ionic compounds Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Chemical Bonding (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Students return to seats. Teacher projects the anchoring phenomenon in full: the Thika Road street lamp (sodium), the jua kali welder's argon shield, the calcium hypochlorite tablet in the Nairobi Water storage tank, and the helium party balloon. Below the image, the unit driving question is displayed: 'Why do some elements explode in water, some glow silently in glass tubes, and others rust on a Kenyan farm — and how do their family properties explain the materials that build, light up, and sustain our daily lives?' Students individually complete a structured CER (Claim–Evidence–Reasoning) worksheet for the FULL anchoring phenomenon, requiring one CER row per chemical family (five rows: alkali metals, alkaline earth metals,	Project the anchoring phenomenon image and driving question. Say: 'This is what we started with eight lessons ago. Today you will write the full scientific explanation — not just one part, all five chemical families, all five Kenyan contexts.' Distribute the CER worksheet. 'You have 15 minutes — work alone. I will collect these at the end of the lesson as your individual formative product.' [WAIT 15 minutes; circulate, reading over shoulders. Do NOT give answers; only prompt with questions.] If a student leaves the 'Noble Gases' row blank, say: 'Think about Lesson 5 and the DQB question from Lesson 7: what explains the GLOW of helium and neon if they never form bonds?' [WAIT 10 seconds.] If a student's 'Reasoning' column is thin, say:	Strategy: Claim–Evidence–Reasoning (CER) Framework for Final Explanation Construction. CER is a disciplinary literacy strategy that scaffolds scientific argumentation by separating the three components of a valid scientific explanation. By requiring students to cite specific lesson evidence (e.g., 'Lesson 6, hibiscus bleaching experiment' or 'Lesson 5, noble gas discharge tube simulation'), the strategy prevents vague generalisations and anchors explanation in empirical evidence. The group comparison step allows peer calibration before public share-out, reducing anxiety and improving precision.	Observable indicator: A complete and adequate CER worksheet contains: five claims (one per family), at least eight total pieces of evidence across five rows (minimum one per family, at least two for alkali metals and halogens), and five reasoning statements that explicitly mention electron configuration or valence electrons. Teacher collects worksheets at end of lesson for written feedback. During cold-call share-out, flag any student whose reasoning column uses purely descriptive language ('because it reacts') without mechanistic explanation ('because it readily loses/gains electrons due to its electron configuration') — schedule a 5-minute follow-up with those students in the next lesson.

	halogens, noble gases, transition metals). Each row requires: a CLAIM (one sentence), at least two pieces of EVIDENCE from the unit (citing the specific lesson/experiment), and REASONING (one to two sentences connecting electron configuration to observed behaviour). Students have 15 minutes to complete the worksheet independently, followed by group comparison (5 minutes) and whole-class share-out via cold-call (8 minutes).	'Connect the number of valence electrons to the behaviour you described — that is the reasoning.' [WAIT 10 seconds.] After 15 minutes: 'Compare your CER with your group for 5 minutes. You may add or refine, but mark any changes in a different colour.' [WAIT 5 minutes.] Cold-call Student 1 for Alkali Metals row: 'Wanjiku, read your CLAIM and one piece of EVIDENCE for alkali metals.' [WAIT 10 seconds.] Cold-call Student 2 for Halogens: 'Musa, read your REASONING for halogens — specifically the equation that supports it.' [WAIT 10 seconds.] Cold-call Student 3 for Noble Gases: 'Achieng, explain the noble gas glow — your full CER row.' [WAIT 10 seconds.] Cold-call Student 4 for Transition Metals: 'Kamau, which Kenyan context did you use for transition metals, and what makes them different from Group 1?' [WAIT 10 seconds.] Cold-call Student 5 for Alkaline Earth Metals: 'Fatuma, what balanced equation did you use as evidence for Group 2 behaviour?' [WAIT 10 seconds.] After each response, ask the class: 'Does anyone want to refine or add to that? Hands up.' Accept two to three additions before moving on. Write the five final equations on the board as students generate them: (1) $2Na + 2H_2O \rightarrow 2NaOH + H_2\uparrow$; (2) $2Mg + O_2 \rightarrow 2MgO$; (3) $Cl_2 + H_2O \rightarrow HCl + HClO$; (4) $Cl_2 + 2NaOH \rightarrow NaCl + NaClO + H_2O$; (5) $4Fe + 3O_2 + 6H_2O \rightarrow 4Fe(OH)_3$ (rust, simplified). Point to the noble gases row: 'No equation for noble gases — why not?' [WAIT 10 seconds.] Elicit: 'Full outer shells — no tendency to gain, lose, or		

		share electrons. That is WHY helium in a party balloon does nothing, and WHY argon protects the jua kali weld without reacting with the hot metal.'		
Phase 4 — Driving Question Board (DQB) Completion 📹 **VIDEO:** [Naming ions and ionic compounds] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Chemical Bonding (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	The class DQB (a large poster or whiteboard section maintained since Lesson 1) is displayed at the front. All sticky-note questions accumulated over eight lessons are visible, colour-coded by lesson. The teacher reads each open question aloud. Students — now working as a whole class — vote by raising hands: 'Can we answer this now?' For questions that receive majority 'yes,' a volunteer student writes the answer directly on a green sticky note and places it over the original question. Special focus is given to the Lesson 7 sticky note: 'We can now explain bleaching — but what about elements that do NOTHING at all?' Students compose a group answer for this question together. The teacher then asks: 'Are there any questions on our DQB that we STILL cannot fully answer?' Any unresolved questions are noted as a bridge to Sub-Strand 2.3 (Chemical Bonding) — particularly questions about WHY noble gases have full outer shells and whether 'full shells' relates to how other elements form bonds.	Stand beside the DQB. 'We have been building this board since Lesson 1. Let us answer every question together.' Read the Lesson 7 sticky note aloud: 'We can now explain bleaching — but what about elements that do NOTHING at all?' Ask: 'Who wants to compose our answer? I need one volunteer to write and three to contribute ideas.' [WAIT 10 seconds for hands.] As the volunteer writes, guide: 'Make sure your answer mentions: electron configuration, full outer shell, and at least one Kenyan example — helium balloon or argon welding shield.' After the answer is posted, say: 'Read every remaining open question to yourself for 60 seconds.' [WAIT 60 seconds.] 'Raise your hand if you believe we can now answer the question about noble gas glow.' Accept four to five student answers, building toward: 'Noble gases have complete outer electron shells. They do not lose, gain, or share electrons. When electricity passes through them in a sealed glass tube, the electrons are temporarily excited to higher energy levels — when they fall back, they emit light of a characteristic colour. No chemical reaction occurs — the noble gas is unchanged.' Write this on the board as a model answer. Then: 'What questions are still partially open?' Expect students to raise questions about HOW atoms form bonds — validate these and write them on a blue 'Next Unit'	Strategy: Collaborative DQB Closure with 'Can We Answer It?' Voting. The public DQB closure ritual makes the growth of collective class knowledge visible and tangible. Voting ('Can we answer this now?') promotes metacognitive monitoring — students must assess their own understanding before committing to an answer. The deliberate preservation of one or two unresolved questions (bridged to Sub-Strand 2.3) models authentic scientific practice: good science generates new questions even as it answers old ones.	Observable indicator: At least 85% of DQB sticky notes receive a green answer note by the end of this phase. The teacher notes which questions cannot be answered — this is diagnostic for both individual students and the class. If fewer than 70% of questions can be answered, this signals that one or more lessons earlier in the unit required insufficient consolidation; the teacher should note this in the Teacher Reflection and plan a short revision activity. The quality of the group-composed noble gas answer is assessed against three criteria: (1) mentions electron configuration, (2) mentions full outer shell / inertness, (3) includes a Kenyan daily-life example.

		sticky note on the DQB: 'To be answered in Sub-Strand 2.3: Chemical Bonding.'		
Phase 5 — Model Building: L1 vs Final Model Comparison and Written Final Explanation 📹 **VIDEO:** [Naming ions and ionic compounds](#) Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Chemical Bonding (Flexbook)](#) Source: CK-12 🔍 Search ARES for similar readings	Students retrieve their most recent model (from Lesson 7 or the Gallery Walk annotations) and place it beside their original Lesson 1 model. Using a two-column 'Model Evolution Record' sheet, they document: (Column 1) What my L1 model said; (Column 2) What my Final model says; (Column 3) The specific evidence/experiment that caused this change. They complete at least four rows (one per chemical family evolution). Students then write a Final Explanation paragraph of 80–120 words that answers the driving question directly, using at minimum: three chemical family names, two balanced equations, two Kenyan contexts, and the phrase 'electron configuration.' The paragraph is written individually in exercise books. Two to three students volunteer to read their paragraph aloud. Teacher synthesises by projecting a 'model answer' paragraph and comparing it live with two student responses, highlighting strengths and one area for growth in each. Finally, teacher explicitly bridges: 'Everything we have explained today — why sodium reacts, why chlorine bleaches, why argon is inert — all comes down to electron configuration and valence electrons. In Sub-Strand 2.3, we will ask: what happens when atoms USE those electrons to form bonds with each other? That is where Chemical Bonding begins.'	Distribute the Model Evolution Record sheet. 'You have 8 minutes to complete at least four rows comparing your L1 model to your final model.' [WAIT 8 minutes; circulate and read entries.] If a student's Column 3 ('evidence that caused change') is blank, prompt: 'Which experiment or video in this unit showed you something your L1 model could not explain? Write that experiment here.' [WAIT 10 seconds.] After 8 minutes: 'Now write your Final Explanation paragraph — 80 to 120 words, alone, no group discussion. You have 7 minutes.' [WAIT 7 minutes.] Call for volunteers: 'Who would like to read their paragraph? I need two brave scientists.' Accept two to three volunteers; after each, ask the class: 'One thing they explained well — hands up.' [WAIT 5 seconds; accept two responses.] Project the model answer paragraph. 'Here is one way to write this explanation. Notice it has three family names, two equations, two Kenyan contexts, and the phrase electron configuration. Check your own paragraph against these four criteria — put a tick beside each one you included.' [WAIT 2 minutes for self-check.] Close: 'In Lesson 1, you could not explain why sodium glows on Thika Road and helium does nothing in a party balloon. Now you can — and the explanation fits on one page. That is the power of chemical families. Next lesson, we go deeper: not just which family an element is in, but HOW its electrons join with another atom's	Strategy: Model Evolution Record + Final Explanation Writing. The three-column Model Evolution Record is a metacognitive tool that forces students to attribute each conceptual change to specific empirical evidence — this is the core epistemic practice of science. The Final Explanation paragraph synthesises across five chemical families, two modalities (text and equations), and multiple Kenyan contexts into a single coherent argument. The live comparison of student paragraphs with the model answer normalises revision and improvement as part of scientific writing, rather than framing the model answer as a 'correct answer' students failed to produce.	Observable indicator: Each student's Final Explanation paragraph is assessed against four criteria using a simple tick-check: (1) names at least three chemical families, (2) includes at least two balanced equations, (3) references at least two Kenyan daily-life contexts (e.g., Thika Road street lamp, jua kali welding, Nairobi Water chlorination, ugali flour starch test, mabati/corrugated iron roofing), (4) uses the phrase 'electron configuration' in a meaningful way. Paragraphs are collected as a written formative assessment product. Students who include all four criteria are considered to have met expectations. Students who include all four AND provide a comparison between two families (e.g., explaining why Group 1 is reactive while Group 18 is inert) are considered to exceed expectations. The Model Evolution Record is retained in student portfolios.

		electrons to build every material around you.'		

D. TEACHER REFLECTION
After delivering this Final Explanation lesson, reflect on the following questions: 1. MODEL EVOLUTION QUALITY: Did students' Model Evolution Records show genuine conceptual change, or were students simply re-copying their current knowledge without linking changes to specific evidence from earlier lessons? If Column 3 ('evidence that caused change') was mostly blank or vague, this suggests that earlier lessons did not sufficiently emphasise the evidence-to-explanation link — plan to make this more explicit in Lessons 3–6 when this unit is taught again. 2. CER WORKSHEET ANALYSIS: After collecting the CER worksheets, sort them into three piles: 'Exceeds' (all 4 criteria met plus family comparisons), 'Meets' (all 4 criteria met), 'Approaches/Below' (fewer than 4 criteria). What percentage of the class falls in each pile? If more than 30% are in 'Approaches/Below,' schedule a 20-minute consolidation activity at the start of the next lesson before beginning Sub-Strand 2.3. 3. DQB CLOSURE RATE: What percentage of DQB questions received a green answer note? If fewer than 80% were answered, identify which lesson's questions remained open — this pinpoints which chemical family was least well understood. Noble gases and transition metals are historically the most under-explained; if these were the open questions, consider adding a dedicated 15-minute consolidation activity for those families early in Sub-Strand 2.3. 4. KENYAN CONTEXT INTEGRATION: Did students spontaneously use Kenyan contexts (Thika Road, Kamukunji jua kali, Nairobi Water, ugali flour, mabati roofing) in their CER worksheets and Final Explanation paragraphs, or did they revert to generic ('sodium reacts with water') statements? High use of Kenyan contexts signals that the unit's contextualisation strategy is working; low use suggests more deliberate teacher modelling of Kenya-relevant language in future lessons. 5. NOBLE GAS GLOW EXPLANATION: This is consistently the hardest concept — the distinction between inertness (no chemical reaction, explained by full outer shell) and luminescence (energy emission from electron excitation, explained by physics of electron energy levels). If students conflated these in Phase 3, consider introducing a brief analogy: 'A jua kali welder who is very skilled (noble gas) does not react to provocation (inert), but when given extra energy (electricity), they briefly jump into action and then return to normal, releasing that energy as light — they are unchanged by the experience.' This analogy bridges chemistry and physics in a Kenyan daily-life frame. 6. BRIDGE TO 2.3: Did the final 'bridge' moment land clearly? Students should leave with a genuine question: 'HOW do valence electrons actually join atoms together?' If students seemed satisfied and closed rather than curious, consider leaving one DQB question intentionally unanswered — for example: 'If sodium loses one electron and chlorine gains one electron, where does that electron GO, and what holds the salt crystal together?' This question perfectly bridges Chemical Families to Chemical Bonding.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Students observed (across Lessons 1–7, reviewed in the Gallery Walk): sodium fizzing and burning violet in water; the orange-yellow glow of a sodium vapour discharge tube (modelling Thika Road street lamps); a hibiscus flower bleaching white in chlorine water (modelling Nairobi Water chlorination); iodine solution turning blue-black on ugali flour paste (connecting to Strand 1.3 food tests); the inertness of noble gases in sealed discharge tubes (modelling the argon welding shield and helium party balloon); and the slow rusting of iron (modelling the jembe/hoe and mabati corrugated iron roof on a Kenyan farm).

What did I learn?	Students learned that elements belong to chemical families based on their electron configuration and position in the periodic table. Alkali metals (Group 1, e.g. sodium) have one valence electron and react vigorously with water, producing metal hydroxides and hydrogen gas. Alkaline earth metals (Group 2, e.g. magnesium, calcium) have two valence electrons and are less reactive than Group 1 but still reactive with oxygen and acids. Halogens (Group 17, e.g. chlorine, iodine) have seven valence electrons, are highly electronegative, and react with water to form bleaching/disinfecting solutions; they show a colour trend down the group (F_2 pale yellow gas → Cl_2 yellow-green gas → Br_2 red-brown liquid → I_2 grey solid). Noble gases (Group 18, e.g. helium, argon, neon) have complete outer shells, are chemically inert, and produce characteristic coloured glows when electricity passes through them in sealed tubes — with no chemical change occurring. Transition metals (e.g. iron, copper, zinc) have variable oxidation states, are hard and dense, conduct electricity, form coloured compounds, and corrode slowly (unlike alkali metals which react explosively). Every observed behaviour traces back to the number and arrangement of valence electrons.
How does this explain the phenomenon?	Students explained that the anchoring phenomenon — sodium street lamps glowing orange on Thika Road, argon shielding a jua kali weld, chlorine compounds purifying Nairobi Water, and helium floating inertly in a party balloon — is fully accounted for by chemical family membership and electron configuration. Sodium glows because it is a Group 1 alkali metal: its single valence electron is excited by electrical energy in the discharge lamp and emits orange-yellow light when it returns to ground state; on Thika Road, thousands of sodium atoms do this simultaneously every night. Argon shields the weld because it is a Group 18 noble gas with a full outer shell (eight electrons), making it chemically inert — it neither reacts with the hot metal nor with atmospheric oxygen, providing a protective blanket with no chemical consequences. Chlorine (Group 17, seven valence electrons) reacts with water via $Cl_2 + H_2O \rightarrow HCl + HClO$; the hypochlorous acid (HClO) produced is a powerful oxidising bleach that destroys the chromophores in the hibiscus petals and kills pathogens in Nairobi Water storage tanks. Helium (Group 18, full outer shell with two electrons) is completely inert and non-toxic, making it the safe choice for party balloons. Iron (a transition metal) corrodes slowly via $4Fe + 3O_2 + 6H_2O \rightarrow 4Fe(OH)_3$ because its partially filled d-orbitals give it moderate reactivity — far less than sodium, but more than noble metals. The periodic table is therefore a map of chemical behaviour: knowing a element's family tells you how it will behave before you ever touch it.

DIFFERENTIATION AND INCLUSION		
Learner Need	**Support Strategies**	**Extension Strategies**
English Language Learners	Provide bilingual glossaries; allow use of first language for initial model drawing.	Write extended explanations of observations; lead group discussions in English.
Students with learning difficulties	Provide structured note frames; pair with a supportive partner during investigations.	Mentor peers; create additional visual models to explain concepts.
Gifted and advanced learners	Access to additional reading materials; challenge questions during DQB.	Lead model-building discussions; research and present extension topics to class.